MY RESEARCH AND WORK

Chromatic fixed point theory and topology in aviation

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LinkedIn
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Slides (Web)

PART 1

Chromatic fixed point theory

Smith theory, Morava K-theory, and real Grassmannians

PART 2

Topology in aviation

Detecting airborne holding in near real time

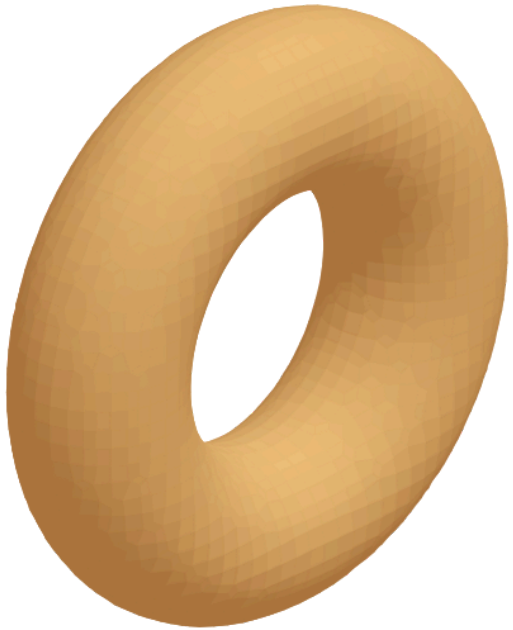
I am a classically trained algebraic topologist.

I have been working at the **FAA** for **5 years** as an **Operations Research Analyst / Mathematician** in the **Performance Analysis** group.

ALGEBRAIC TOPOLOGY

The study of shapes under continuous deformation

Stretch and bend, but never tear or glue.



ALGEBRAIC TOPOLOGY

The study of shapes under continuous deformation

Stretch and bend, but never tear or glue.

Take a donut. Push down to form the cup.

Then bend the rest around to make the handle.

Both have exactly one hole.

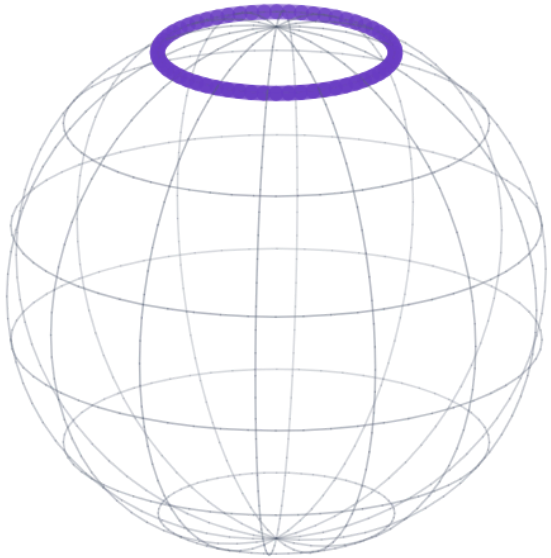


ALGEBRAIC TOPOLOGY

Attach an algebraic gadget to each space

A number, a group, a vector space: something that tells spaces apart.

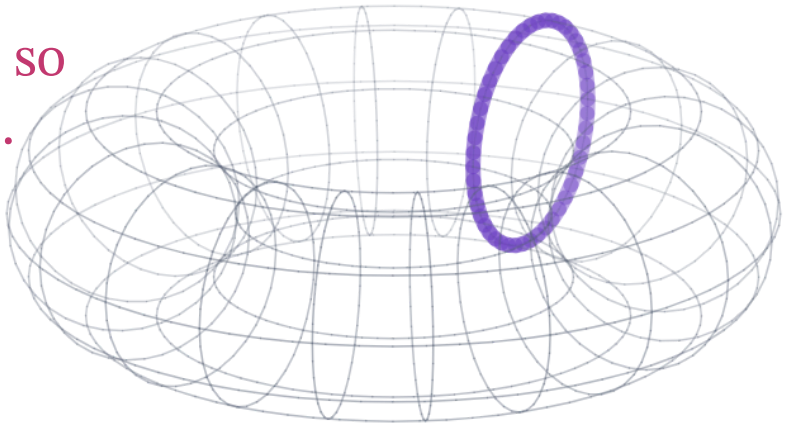
For example, homology attaches a vector space that counts holes.



this loop shrinks to a point

$$\dim H_1(S^2) = 0$$

Different gadgets, so different spaces.



this loop cannot shrink

$$\dim H_1(T^2) = 2$$

A primer on mod 2

Mod 2 arithmetic, written $\mathbb{F}_2$, has just two numbers, **0** and **1**, with one rule: $1 + 1 = 0$.

In logic terms: addition is XOR and multiplication is AND.

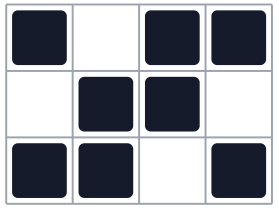
We can do **linear algebra** over $\mathbb{F}_2$: vectors and matrices are built from 0s and 1s, and we add and multiply them with these two tables.

In this talk we will work mod 2. Homology will be **mod 2 homology**.

$+$	0	1
0	0	1
1	1	0

$\times$	0	1
0	0	0
1	0	1

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \iff$$



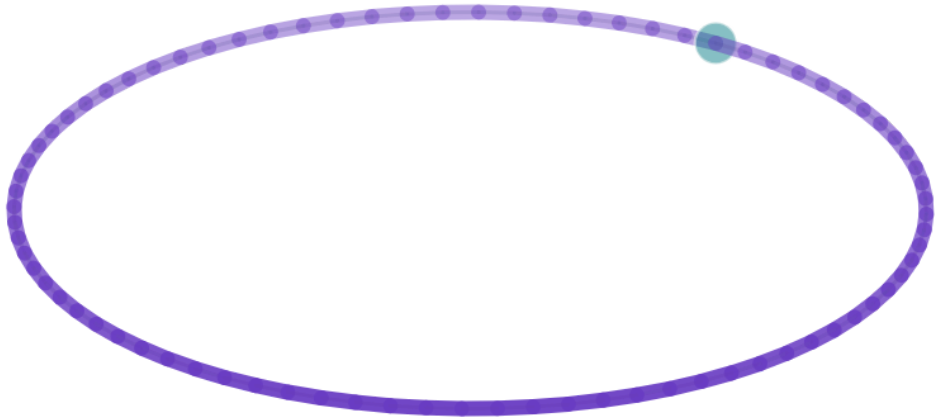
A matrix over $\mathbb{F}_2$, and the same matrix drawn: a filled cell is 1, a blank cell is 0.

HOMOLOGY

Homology studies n-dimensional holes

dim $H_n(X)$ counts the n-dimensional holes of X.

Circle | Sphere | Torus | Two circles | Disk



$\dim H_0(X) = 1$ pieces

$\dim H_1(X) = 1$ loops

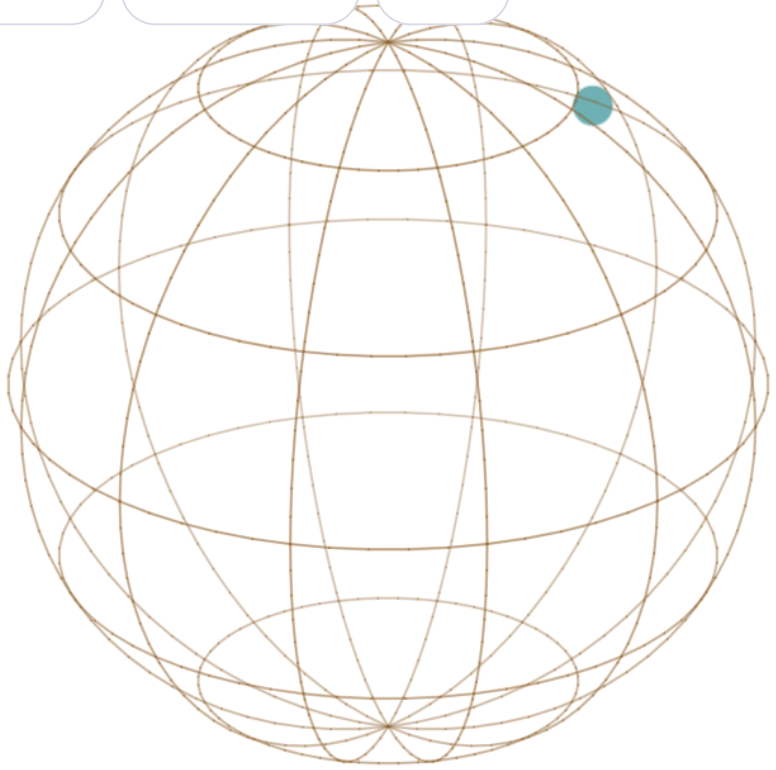
$\dim H_2(X) = 0$ voids

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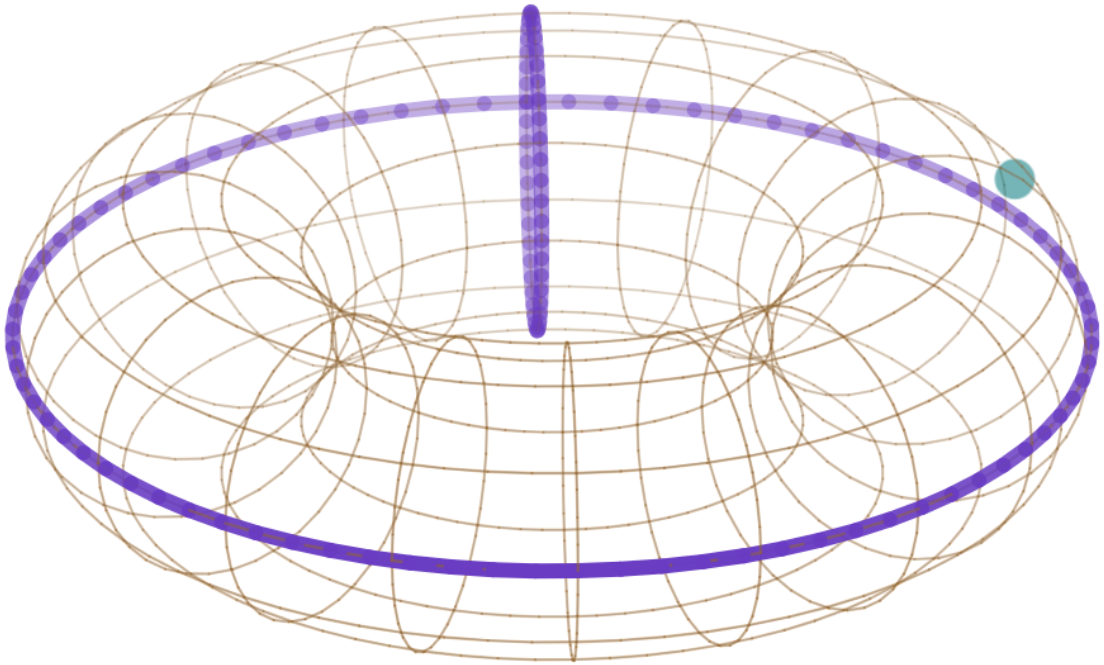
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Circle | Sphere | **Torus** | Two circles | Disk



$\dim H_0(X) = 1$ pieces

$\dim H_1(X) = 2$ loops

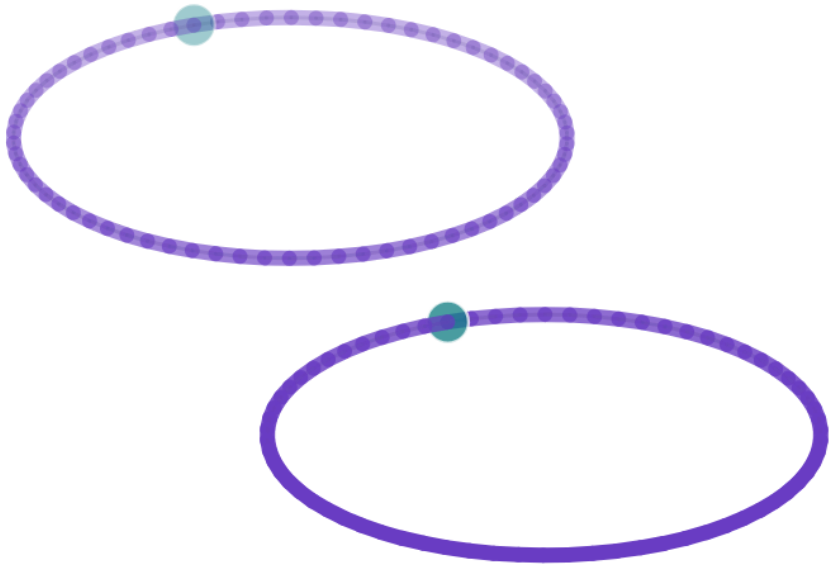
$\dim H_2(X) = 1$ voids

HOMOLOGY

Homology studies n-dimensional holes

dim $H_n(X)$ counts the n-dimensional holes of X.

Circle | Sphere | Torus | **Two circles** | Disk



$\dim H_0(X) = 2$ pieces

$\dim H_1(X) = 2$ loops

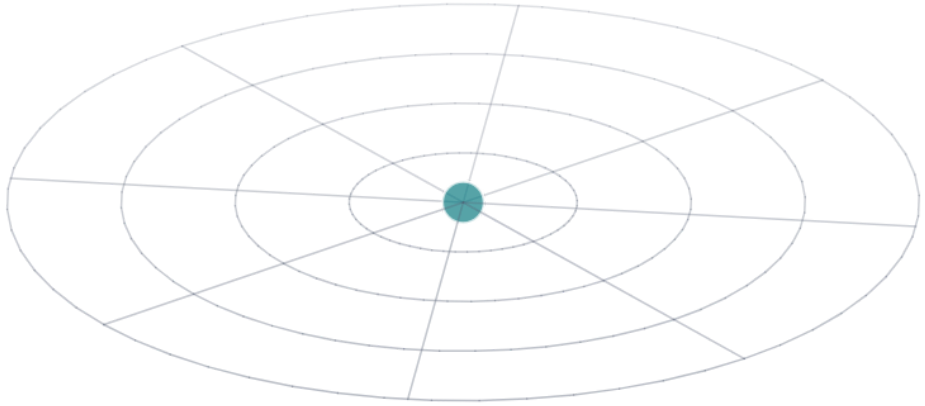
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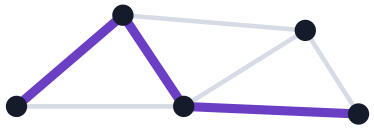
$\dim H_2(X) = 0$ voids

A disk is contractible, so it is acyclic: its homology is that of a point.

HOMOLOGY

Homology = cycles modulo boundaries

Chain



A k**-chain** is a formal sum of k-dimensional pieces: points, edges, faces for $k = 0, 1, 2$.

Mod 2: each piece is in the sum or not. These sums form C_k.

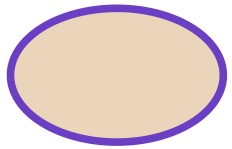
Cycle



A **cycle** is a chain with no loose ends: $\partial c = 0$.

A path has two endpoints, so it is not a cycle.

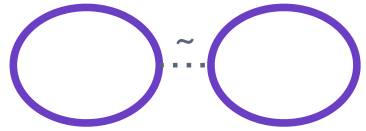
Boundary



A **boundary** is the edge of something one dimension higher: $c = \partial d$.

Every boundary is a cycle: the edge of a disk has no edge.

Homology



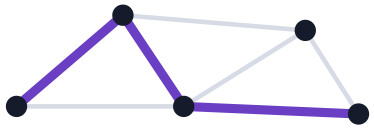
Cycles that are not boundaries, counting two as **the same** if they differ by a boundary.

That counts the holes.

HOMOLOGY

Homology = cycles modulo boundaries

Chain



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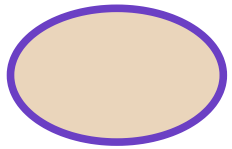
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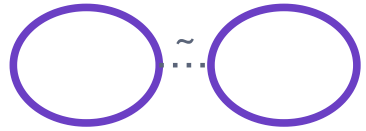
Boundary



A **boundary** is the edge of something one dimension higher: $c = \partial d$.

Every boundary is a cycle: the edge of a disk has no edge.

Homology



Cycles that are not boundaries, counting two as **the same** if they differ by a boundary.

That counts the holes.

$$\cdots \longrightarrow C_{k+1} \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \longrightarrow \cdots \qquad \partial_k \circ \partial_{k+1} = 0$$

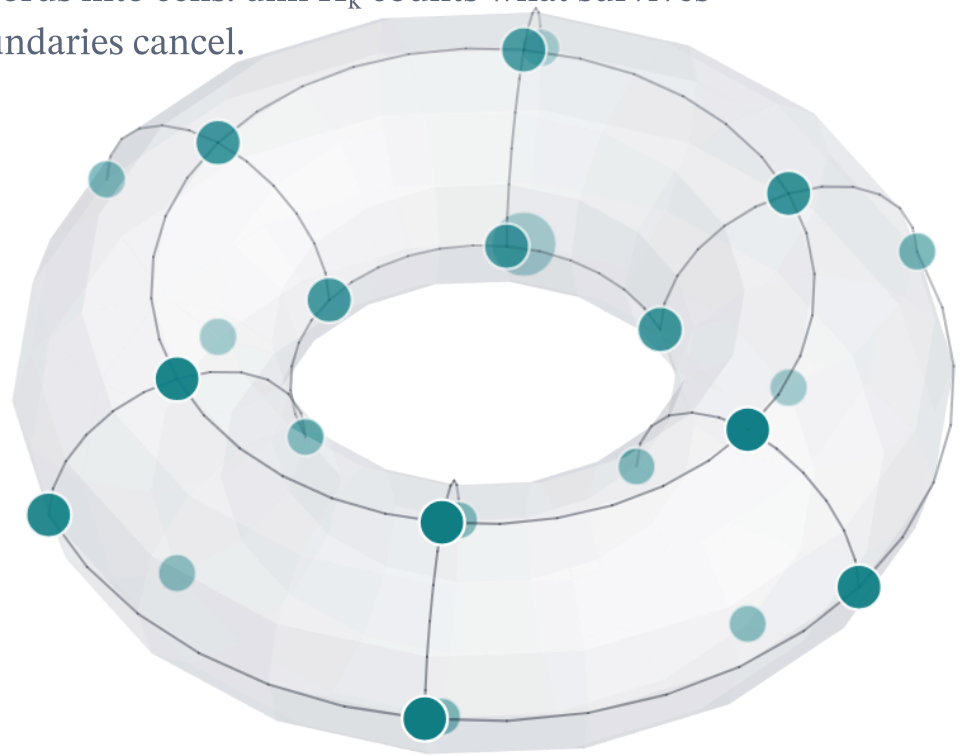
$$H_k = \frac{\ker \partial_k}{\operatorname{im} \partial_{k+1}} \qquad \text{(cycles modulo boundaries)}$$



HOMOLOGY

The same computation, as linear algebra

Cut the torus into cells. dim H_k counts what survives after boundaries cancel.



Points | Edges | Faces

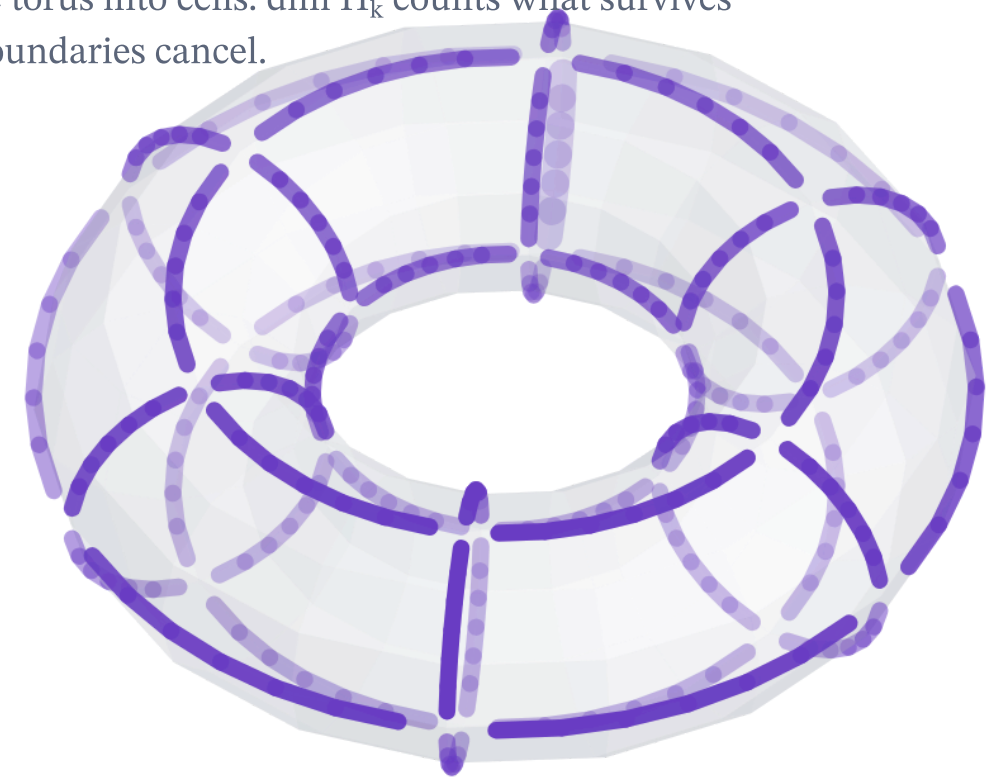
24 points (0-cells)

> **0-cells: points.** One at every corner of the grid: $6 \times 4 = 24$.

HOMOLOGY

The same computation, as linear algebra

Cut the torus into cells. dim H_k counts what survives after boundaries cancel.



Points | Edges | Faces

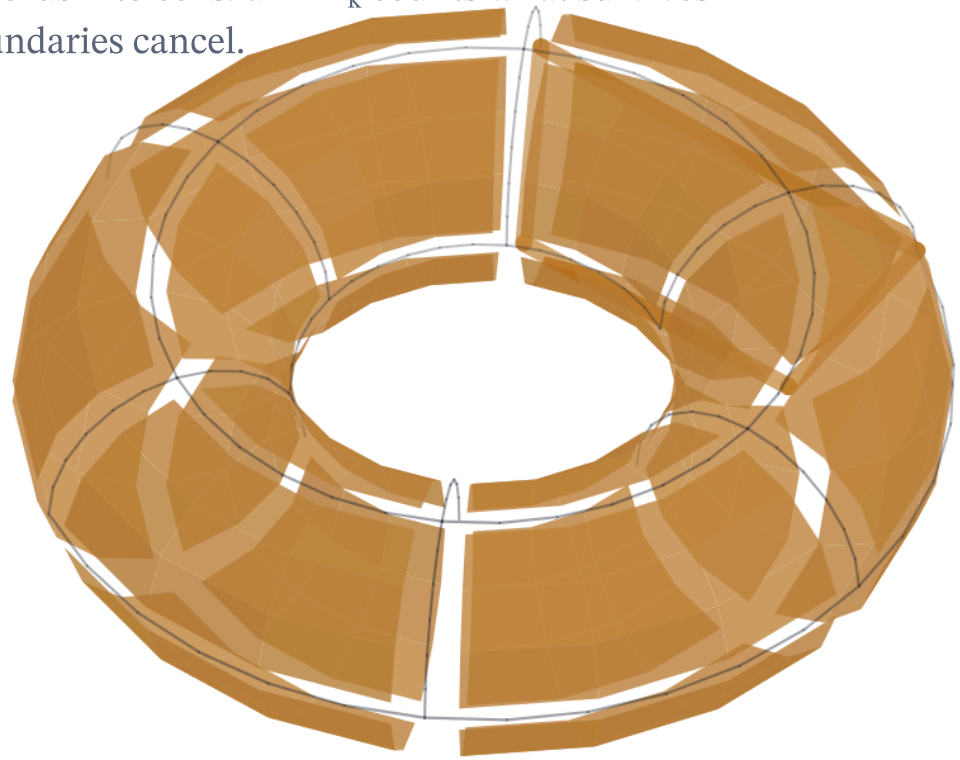
48 edges (1-cells)

> **1-cells: edges.** 24 run around the hole and 24 run around the tube: 48 in all.

HOMOLOGY

The same computation, as linear algebra

Cut the torus into cells. dim H_k counts what survives after boundaries cancel.



Points | Edges | Faces

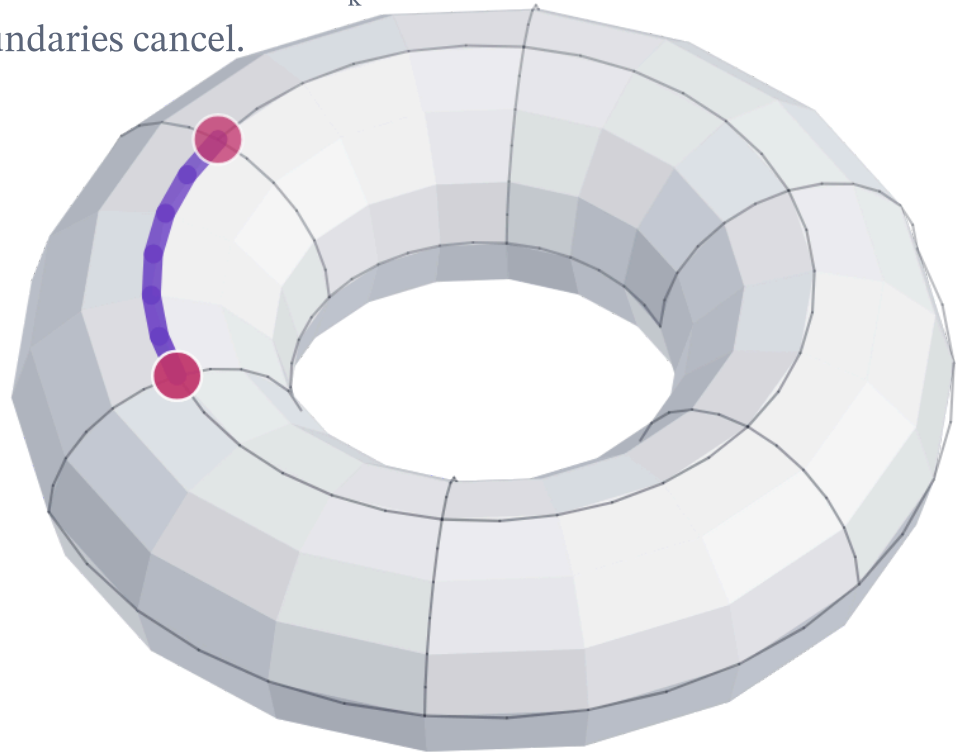
24 faces (2-cells)

> **2-cells: faces.** One square for each grid cell: 6 × 4 = 24.

HOMOLOGY

The same computation, as linear algebra

Cut the torus into cells. dim H_k counts what survives after boundaries cancel.



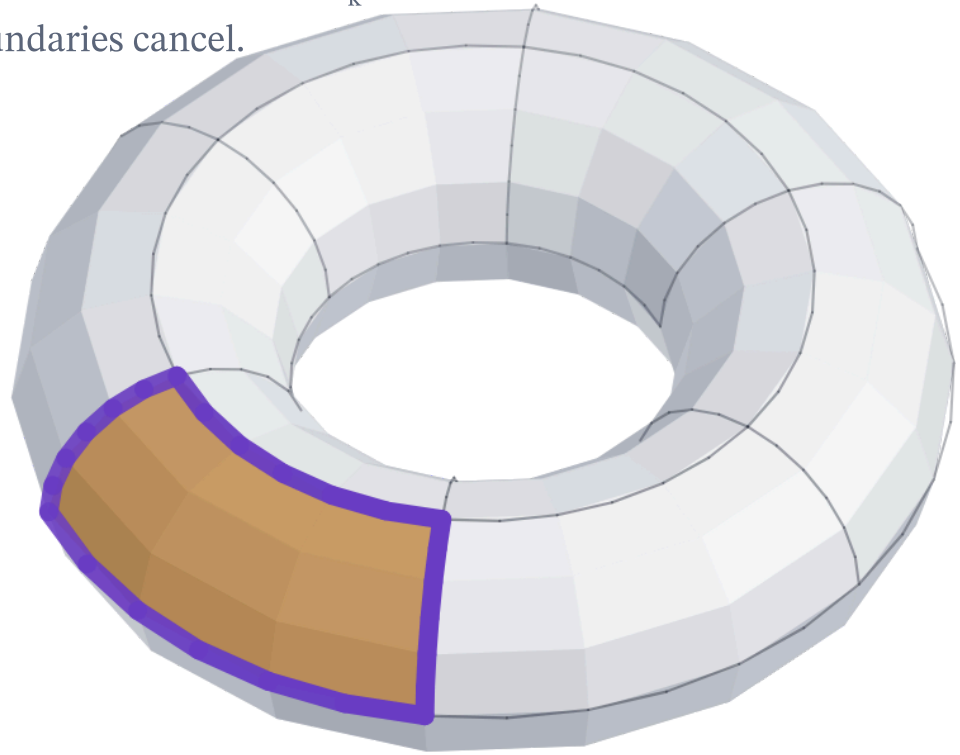
Points | Edges | Faces

An edge. $\partial(\text{edge})$ = its **2 endpoints**, in red.

HOMOLOGY

The same computation, as linear algebra

Cut the torus into cells. dim H_k counts what survives after boundaries cancel.



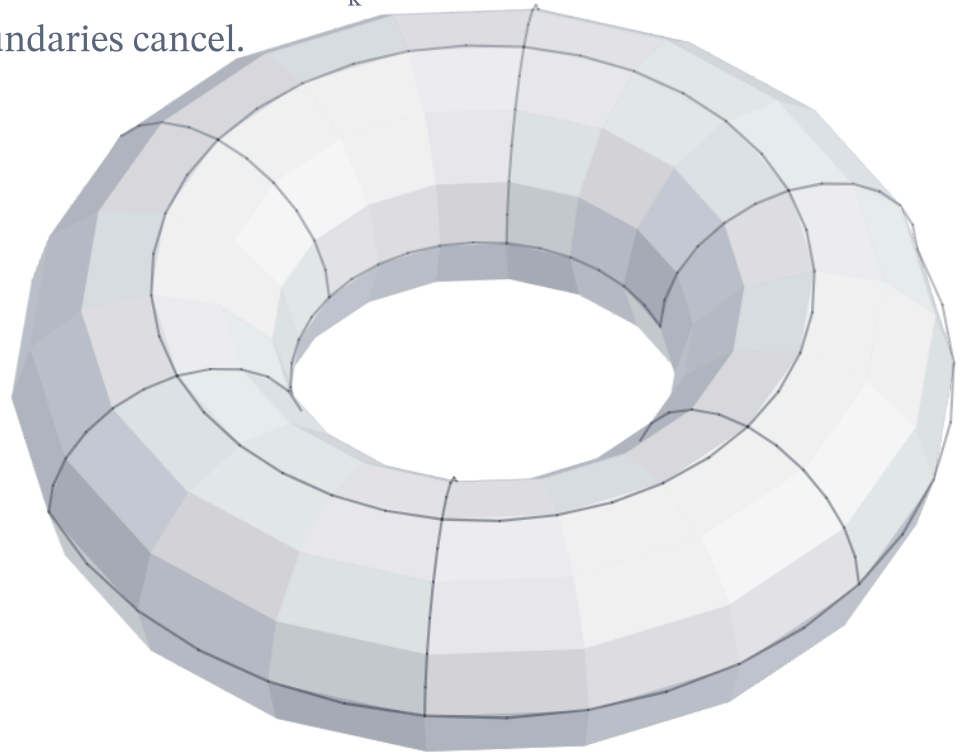
Points | Edges | Faces

A face. $\partial(\text{face}) =$ its **4 edges**, drawn in violet.

HOMOLOGY

The same computation, as linear algebra

Cut the torus into cells. dim H_k counts what survives after boundaries cancel.

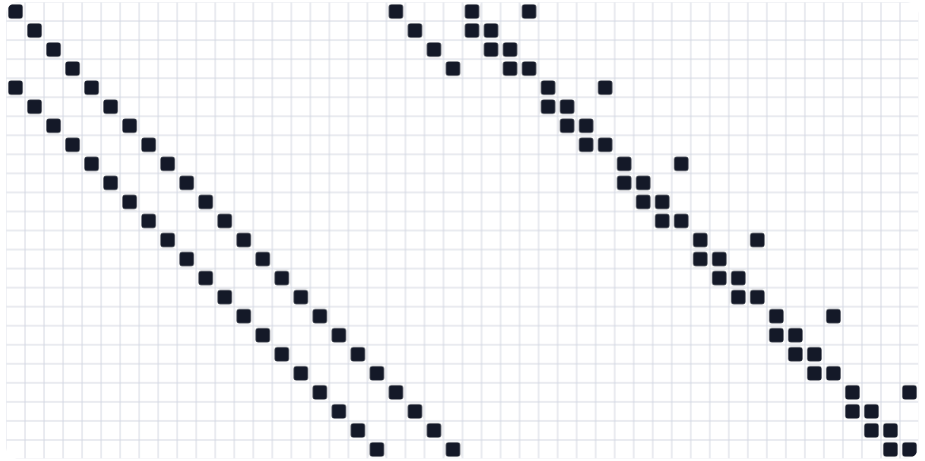


C_2 24 $\xrightarrow{\partial_2}$ C_1 48 $\xrightarrow{\partial_1}$ C_0 24

not cycles · cycles · boundaries · homology

$\partial_1 : C_1 \to C_0$, a 24×48 matrix over $\mathbb{F}_2$

Maps each edge to its two endpoints. Columns are edges, rows are points.



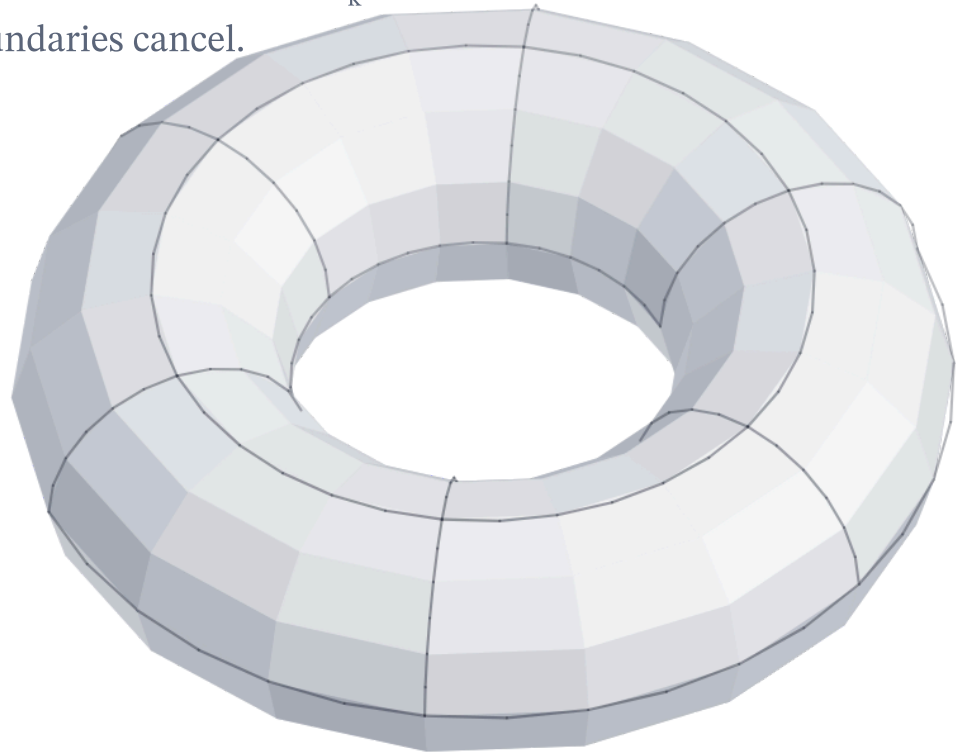
Row Reduce | Step | Result | Reset

We can calculate the dimensions of the cycles and boundaries using elementary linear algebra, by row reduction.

HOMOLOGY

The same computation, as linear algebra

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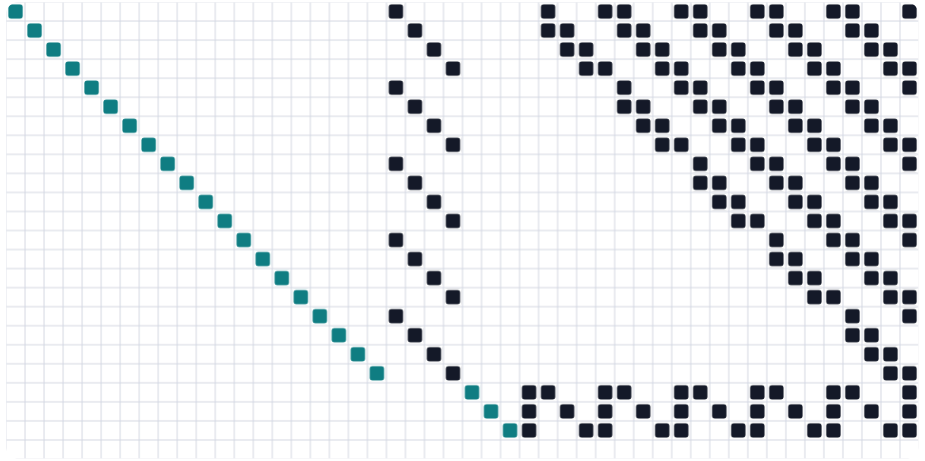




not cycles · cycles · boundaries · homology

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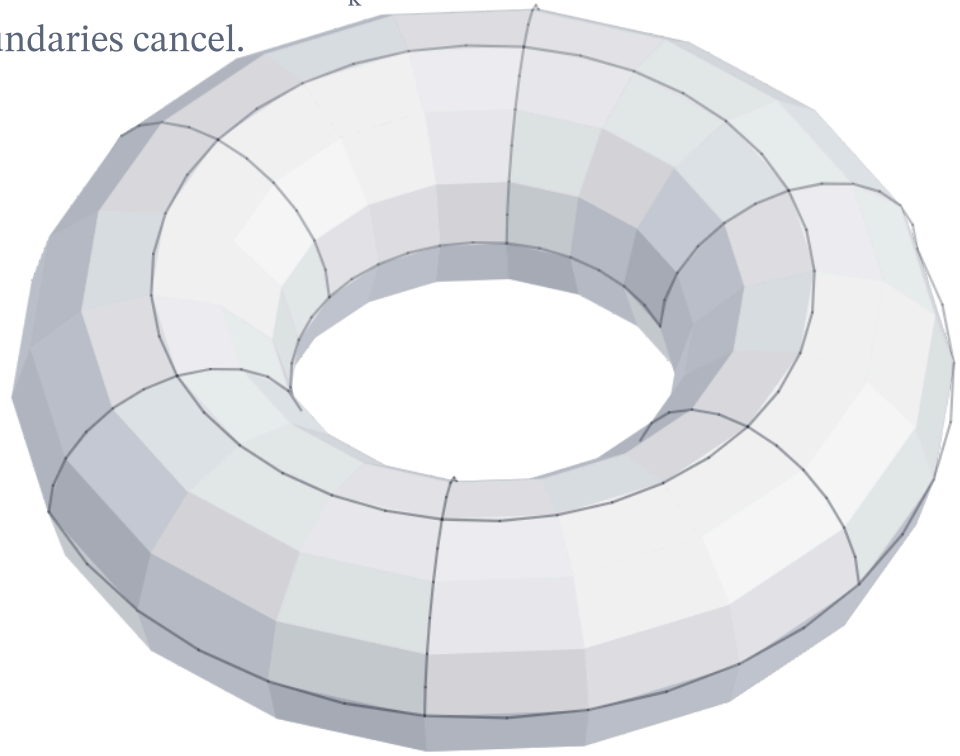
rank = **23**

Done: rank **23**. The 25 free columns give the kernel.

HOMOLOGY

The same computation, as linear algebra

Cut the torus into cells. dim H_k counts what survives after boundaries cancel.

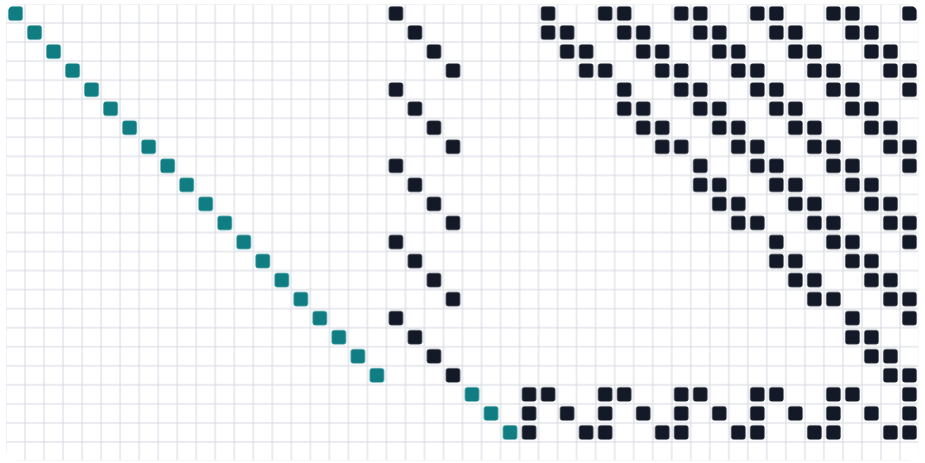




not cycles · cycles · boundaries · homology

$\partial_1 : C_1 \to C_0$, a 24×48 matrix over $\mathbb{F}_2$

Each free column gives one vector x with $\partial_1 x = 0$: a cycle.



◀ Prev · Next ▶ · Pause

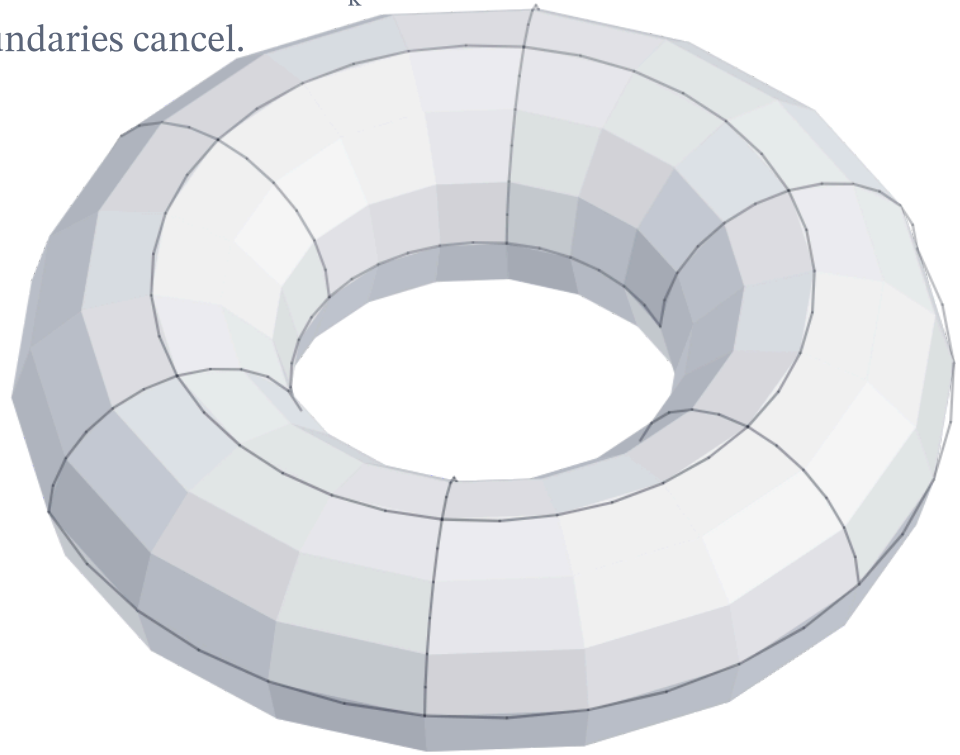
25 free columns = **25** basis cycles

cycles: 48 − 23 = 25. Showing cycle **4** of 25 on the torus.

HOMOLOGY

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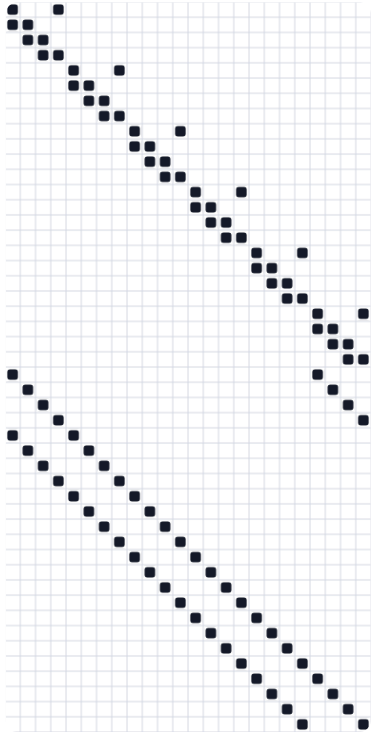




not cycles · cycles · boundaries · homology

$\partial_2 : C_2 \rightarrow C_1$, a 48×24 matrix over $\mathbb{F}_2$

Maps each face to its four edges. Columns are faces, rows are edges.



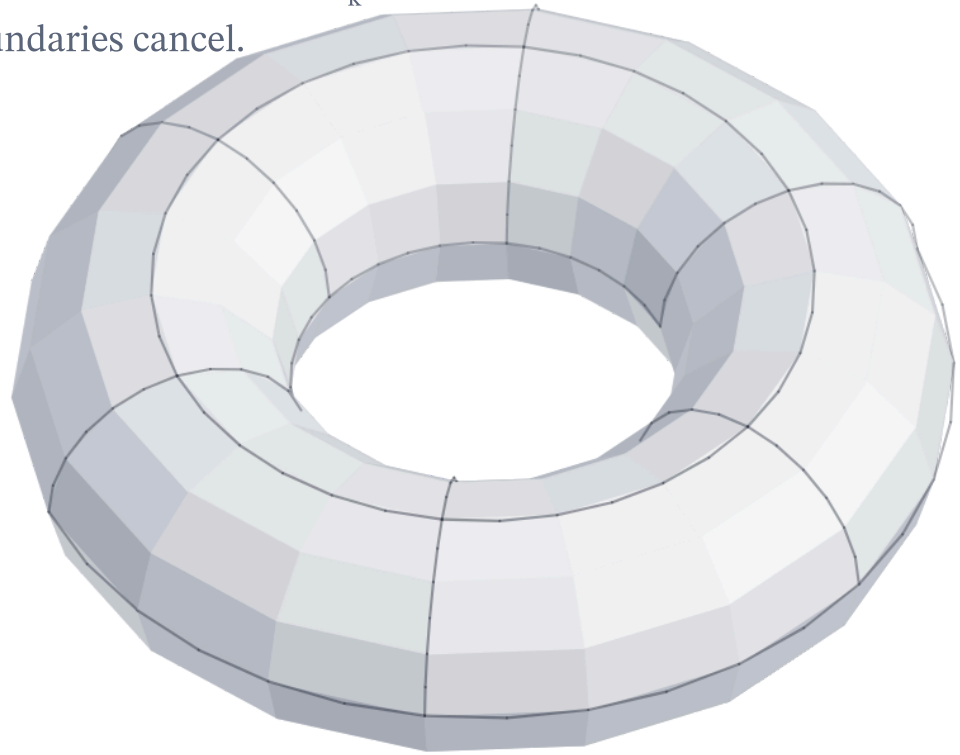
Row Reduce | Step | Result | Reset

We can calculate the dimensions of the cycles and boundaries using elementary linear algebra, by row reduction.

HOMOLOGY

The same computation, as linear algebra

Cut the torus into cells. dim H_k counts what survives after boundaries cancel.

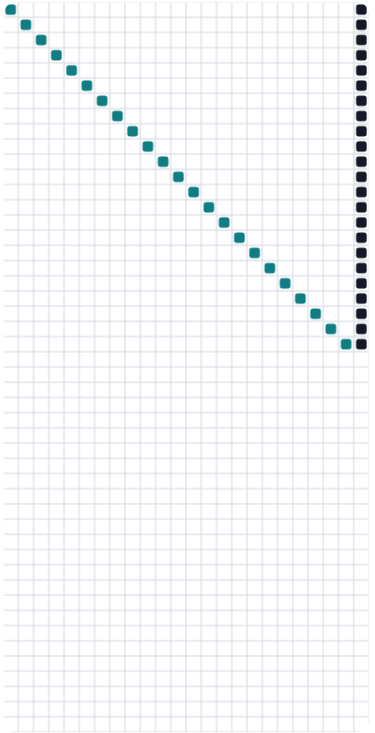




not cycles · cycles · boundaries · homology

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Row Reduce | Step | Result | Reset

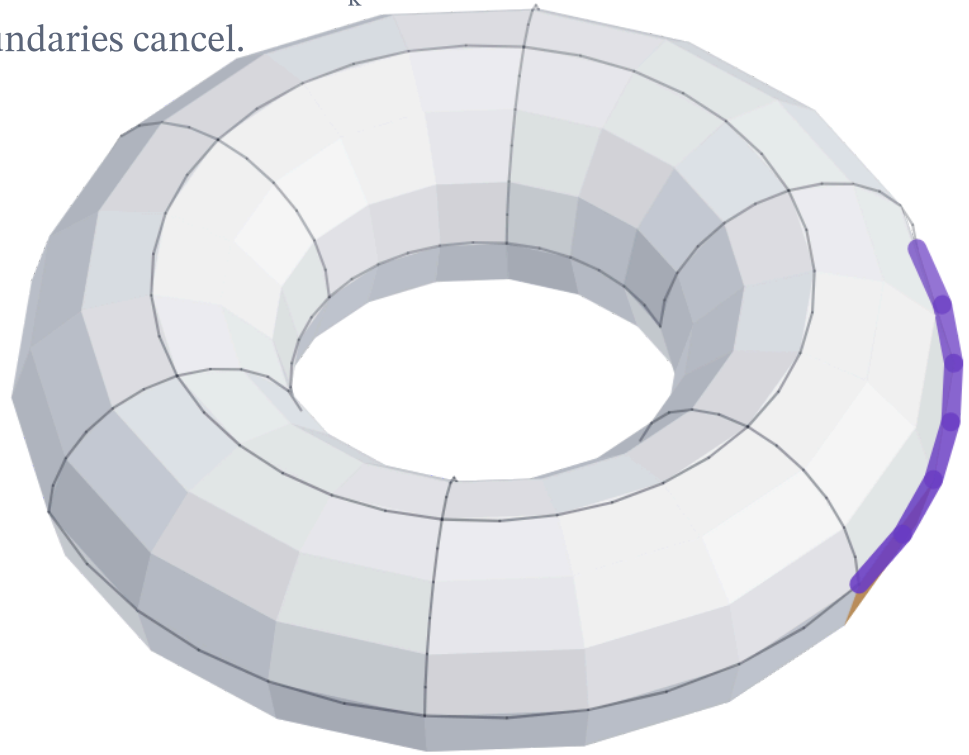
rank = **23**

Done: rank **23**. The 23 pivot columns span the image.

HOMOLOGY

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Cut the torus into cells. dim H_k counts what survives after boundaries cancel.

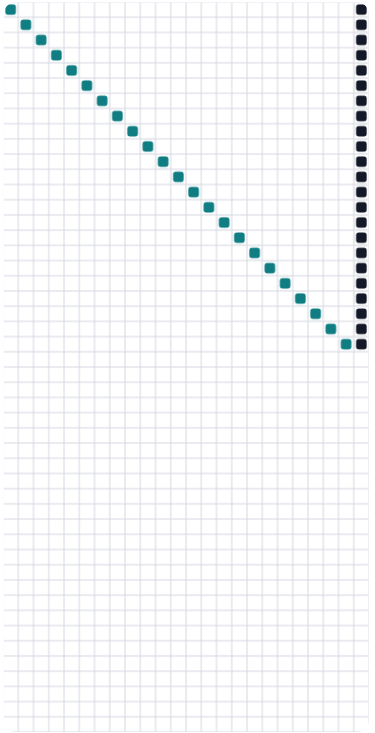




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$\partial_2 : C_2 \to C_1$, a 48×24 matrix over $\mathbb{F}_2$

The pivot columns of ∂_2 are independent boundaries: each is the edge of a face.



◀ Prev · Next ▶ · Pause

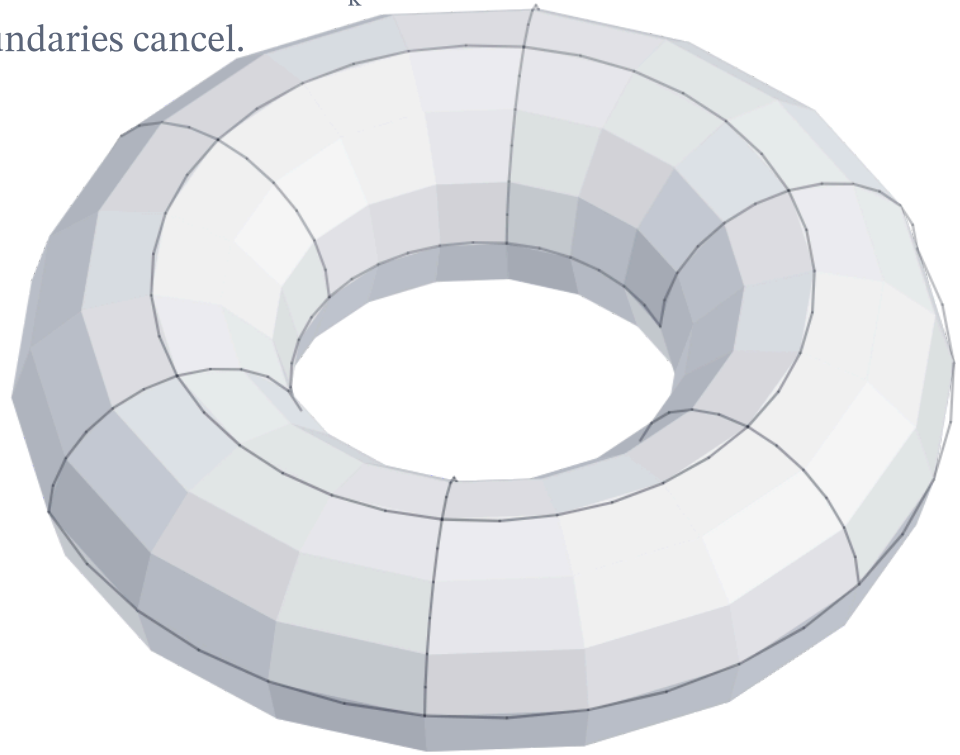
23 pivot columns = **23** independent boundaries

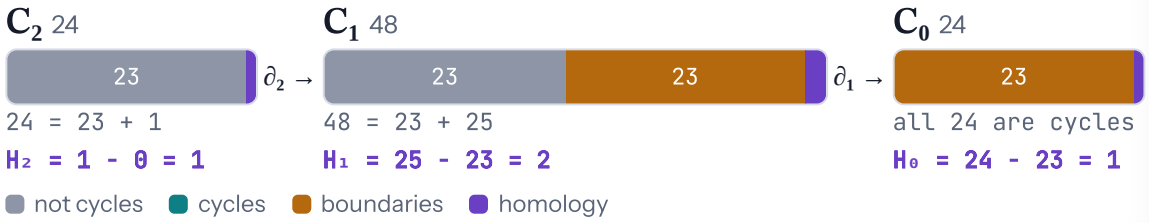
boundaries: rank ∂_2 = 23. Showing the boundary of face **4** on the torus.

HOMOLOGY

The same computation, as linear algebra

Cut the torus into cells. dim H_k counts what survives after boundaries cancel.





Putting it together

Every number in the diagram below comes from row reducing the two matrices.

$\dim H_0(X) = 1$ pieces

$\dim H_1(X) = 2$ loops

$\dim H_2(X) = 1$ voids

$\dim H_k$ = cycles − boundaries: one piece, two independent loops, one void. This is the torus.

GROUPS

What is a group?

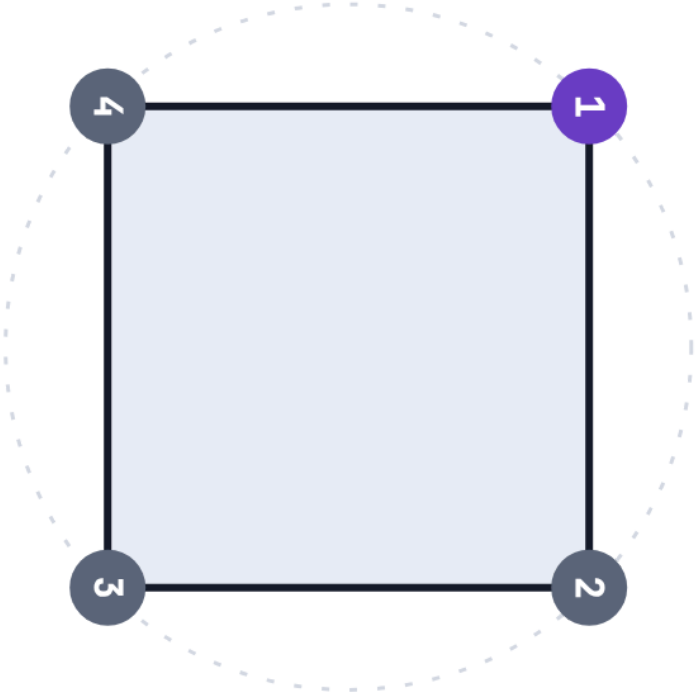
A **group** is a set G with a way to multiply two elements so that

- multiplication is associative,
- there is an identity 1,
- every element has an inverse.

Example. $C_4 = \{1, g, g^2, g^3\}$ with $g^4 = 1$. Here g is a quarter turn of a square.

"Groups are meant to act on things."

g = quarter turn



1 g g^2 g^3 Apply g

GROUP ACTIONS

A group acting on a space

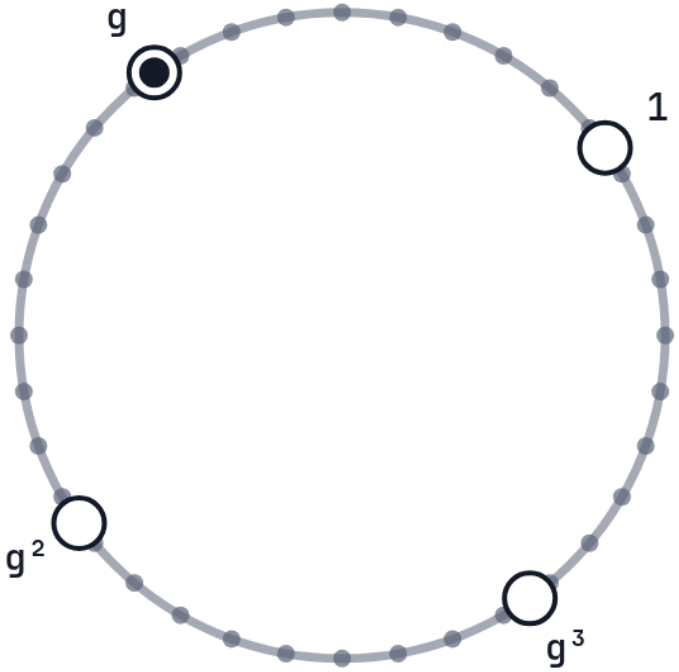
G **acts** on a space X when each $g \in G$ moves the points of X: $x \mapsto g \cdot x$.

Two rules:

- doing nothing does nothing: $1 \cdot x = x$,
- moving twice is moving by the product: $(gh) \cdot x = g \cdot (h \cdot x)$.

Example. C_4 acts on a circle by quarter turns. The **orbit** of a point is where it goes: $x,\ gx,\ g^2x,\ g^3x$.

Quarter turns | Reflection



GROUP ACTIONS

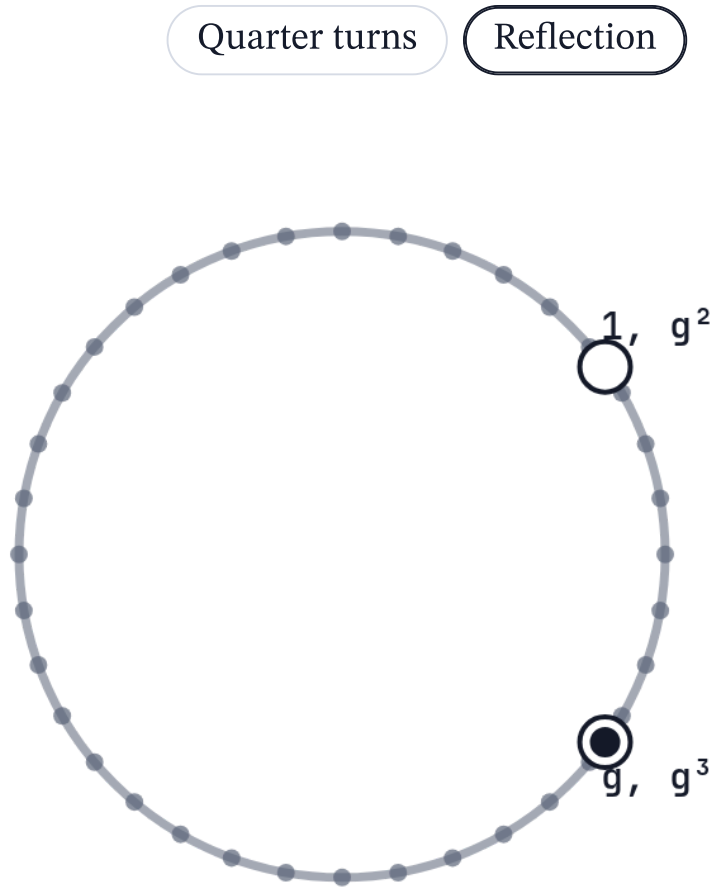
A group acting on a space

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Two rules:

- doing nothing does nothing: $1 \cdot x = x$,
- moving twice is moving by the product: $(gh) \cdot x = g \cdot (h \cdot x)$.

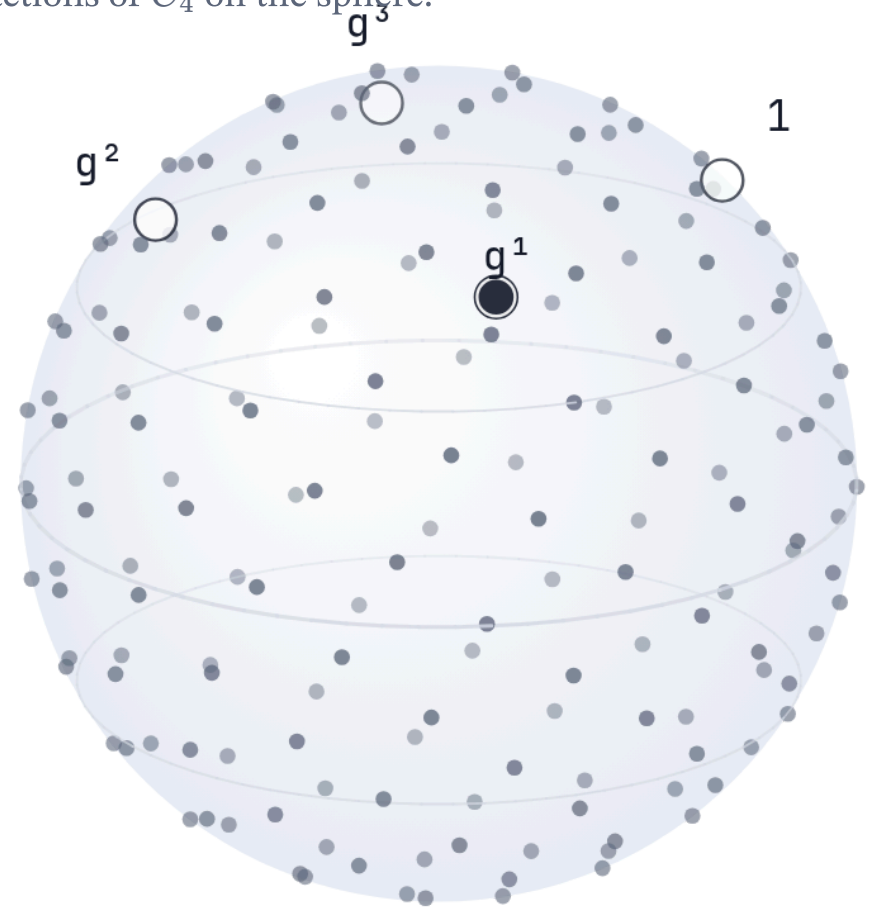
Example. C_4 acts on a circle by quarter turns. The **orbit** of a point is where it goes: $x,\ gx,\ g^2x,\ g^3x$.



GROUP ACTIONS

The same group acting on a sphere

The same group can act on the same space in different ways: here are four actions of C_4 on the sphere.



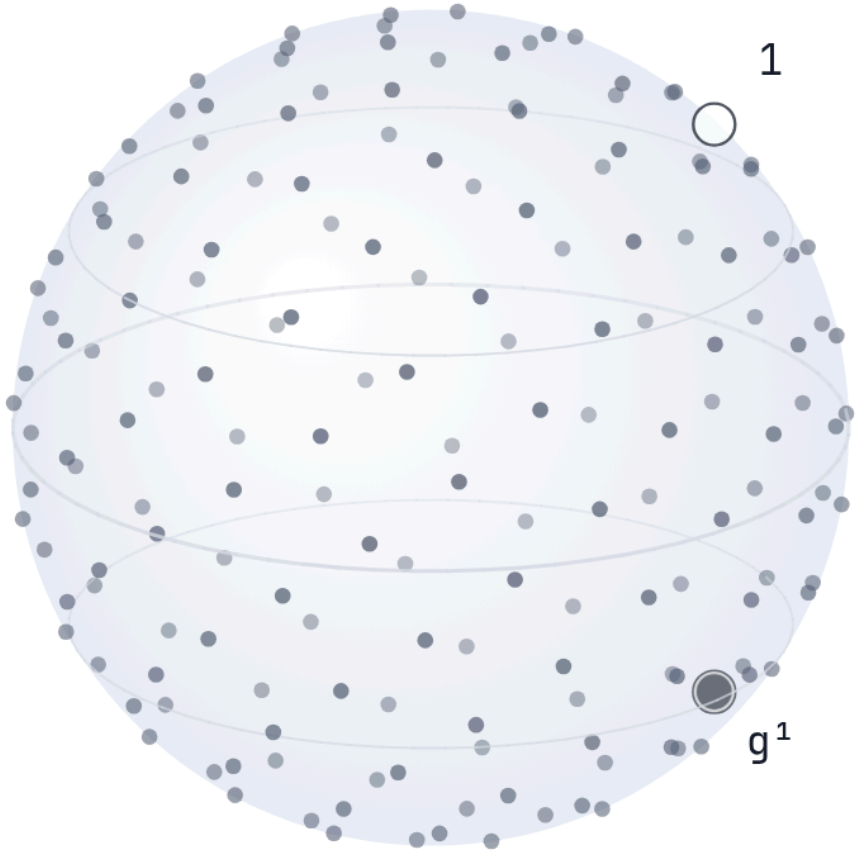
Quarter turn | Reflection | Turn and flip | Antipodal map

A quarter turn about the vertical axis.

GROUP ACTIONS

The same group acting on a sphere

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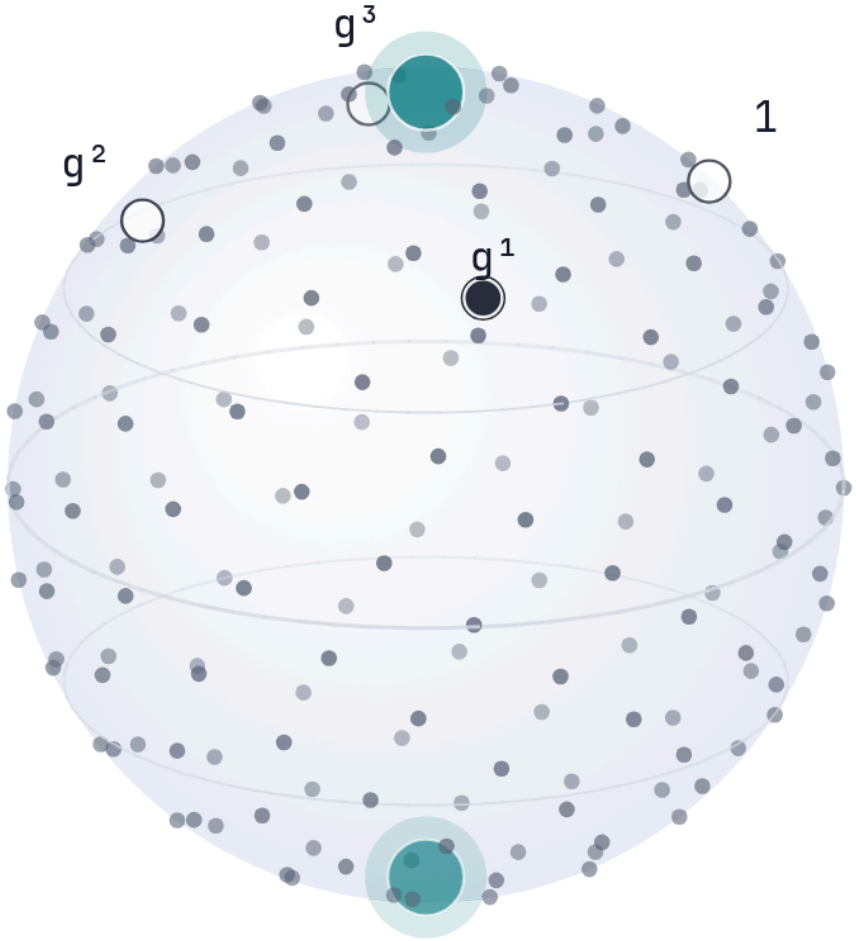
Quarter turn | Reflection | Turn and flip | Antipodal map

A reflection across the equator.

FIXED POINTS

The points that do not move

$X^G = \{x \in X : gx = x \text{ for all } g \in G\}$



$G = C_4$ again, acting on the sphere in four different ways:

Quarter turn | Reflection | Turn and flip | Antipodal map

A quarter turn about the vertical axis.

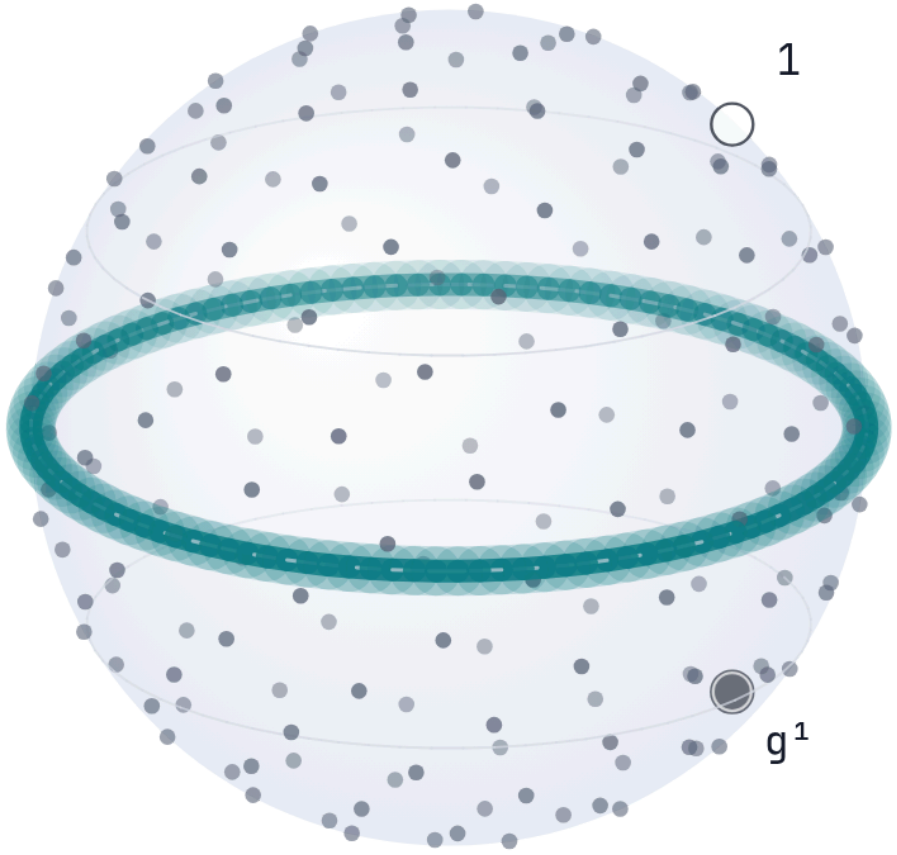
X^G = the two poles

Every point except the two poles is moved. The poles sit on the rotation axis.

FIXED POINTS

The points that do not move

$X^G = \{x \in X : gx = x \text{ for all } g \in G\}$



$G = C_4$ again, acting on the sphere in four different ways:

Quarter turn | **Reflection** | Turn and flip | Antipodal map

A reflection across the equator.

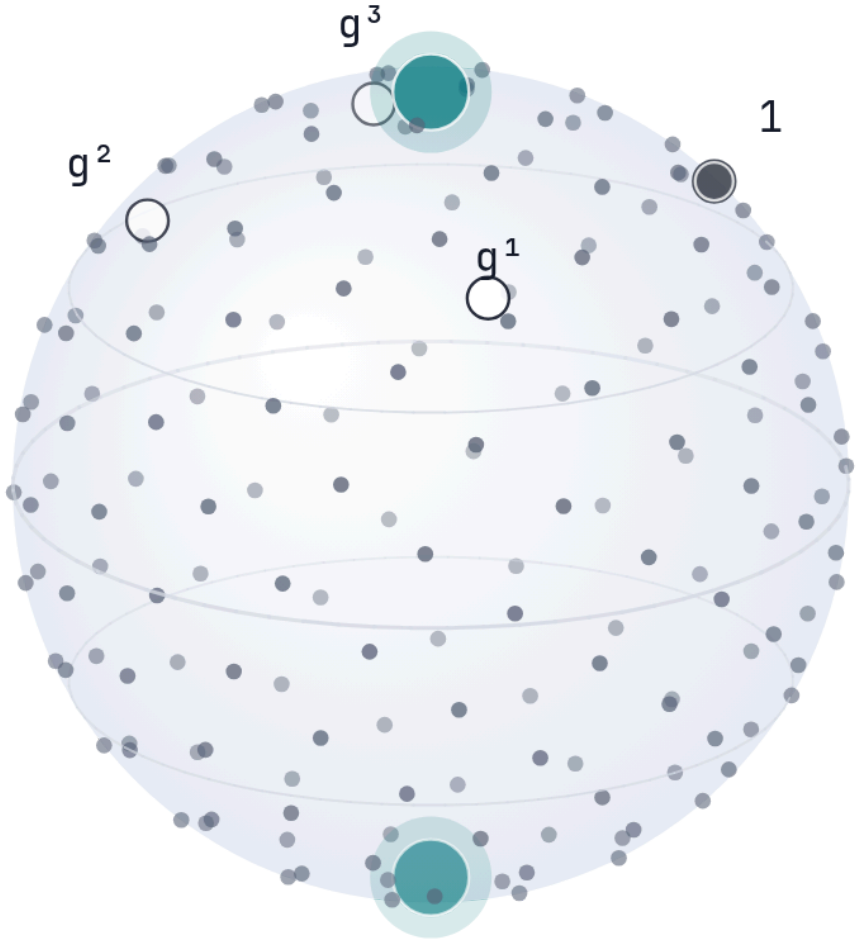
X^G = the equator, a circle

Every point except the equator is moved.

FIXED POINTS

The points that do not move

$X^G = \{x \in X : gx = x \text{ for all } g \in G\}$



$G = C_4$ again, acting on the sphere in four different ways:

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A quarter turn about the vertical axis.

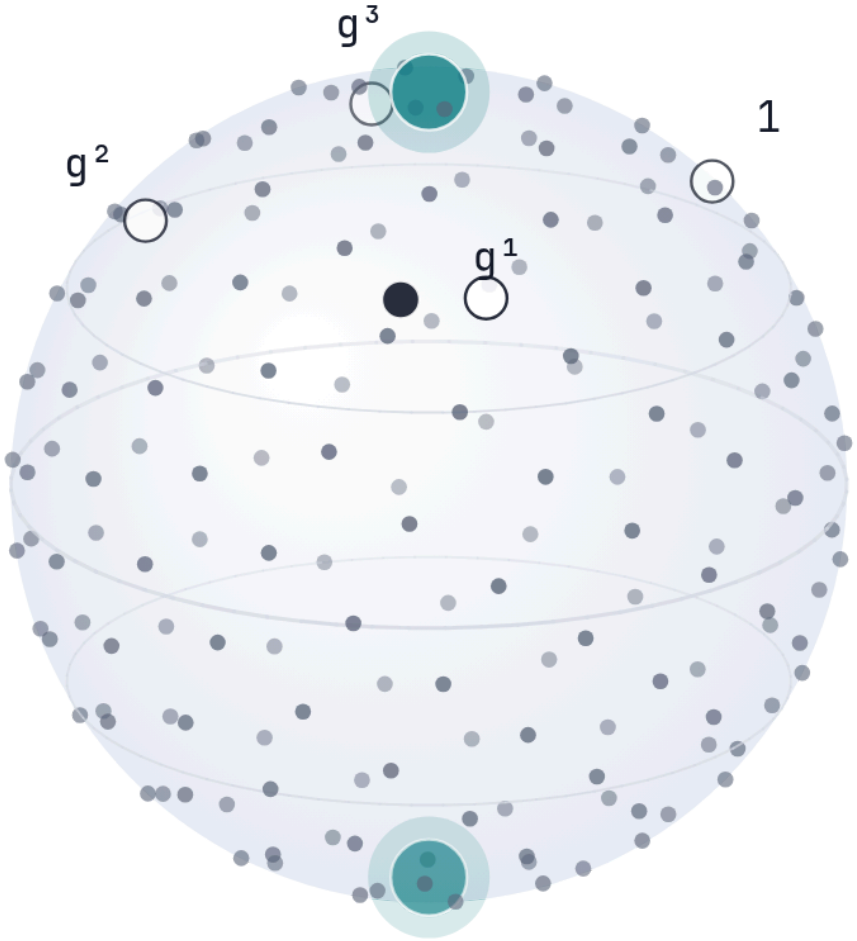
$C_2 = \{1, g^2\} \subset C_4$

The group C_2 lives inside C_4 as a subgroup, generated by g^2 (g done twice). It has fewer elements, so fewer conditions: it can fix more. What does C_2 alone fix?

FIXED POINTS

The points that do not move

$X^G = \{x \in X : gx = x \text{ for all } g \in G\}$



$G = C_4$ again, acting on the sphere in four different ways:

Quarter turn | Reflection | Turn and flip | Antipodal map

A quarter turn about the vertical axis.

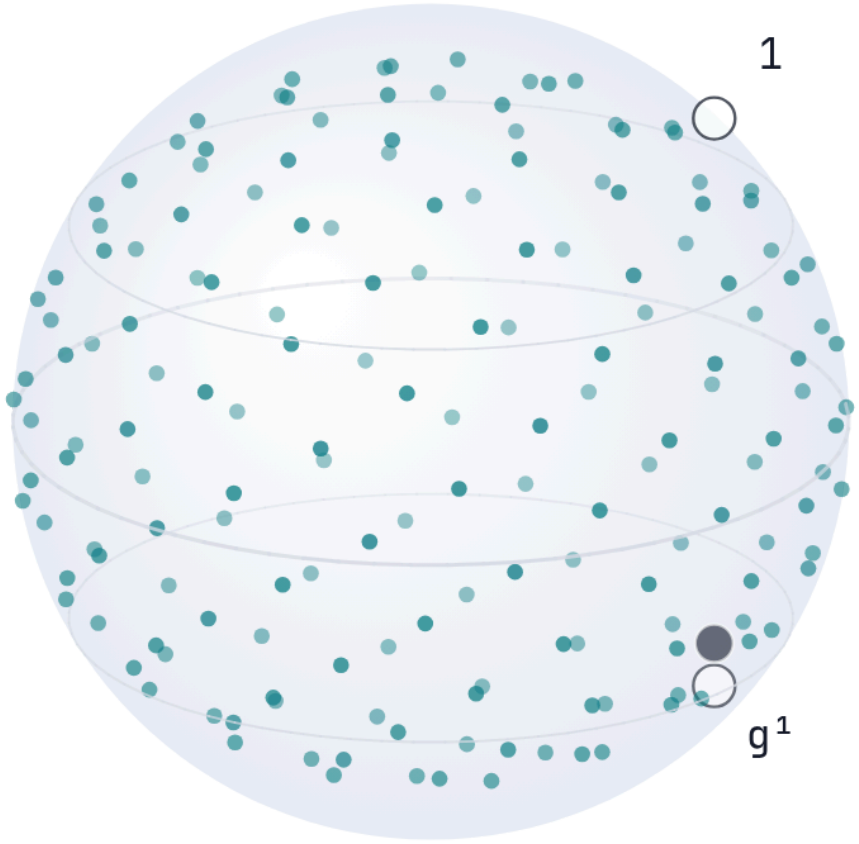
X^{C_2} = the two poles

No change: g^2 fixes exactly the same points as g (the two poles).

FIXED POINTS

The points that do not move

$X^G = \{x \in X : gx = x \text{ for all } g \in G\}$



$G = C_4$ again, acting on the sphere in four different ways:

Quarter turn | **Reflection** | Turn and flip | Antipodal map

A reflection across the equator.

X^{C_2} = all of S^2

Fewer conditions, so more fixed points: the fixed set grows from the equator, a circle to all of S^2.

Fixed by all of C_4 | **Fixed by only $\{1, g^2\}$**

SMITH THEORY

Homology controls fixed points

If a 2-group G acts on a nice space X, the fixed set X^G cannot be arbitrary. Its mod 2 homology is controlled by the mod 2 homology of X.

Acyclic goes to acyclic

$$\tilde{H}_*(X) = 0 \;\Rightarrow\; \tilde{H}_*(X^G) = 0$$

Smith, 1938 to 1941. If X has the homology of a point, so does X^G. Next slide.

Size goes down

$$\dim H_*(X^G) \leq \dim H_*(X)$$

Floyd, 1952. The fixed set has at most as much total homology.

SMITH THEORY

Smith's theorem for acyclic spaces

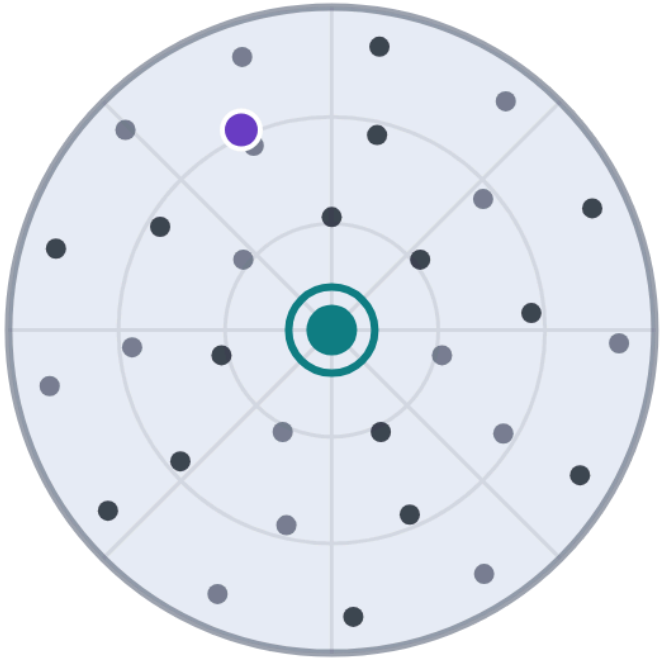
SMITH, 1938 TO 1941

A finite 2-group G acts on a sufficiently nice space X. Then

$$\tilde{H}_*(X) = 0 \quad \Longrightarrow \quad \tilde{H}_*(X^G) = 0.$$

In particular $X^G \neq \emptyset$.

Quarter turn | Reflection



Example. $G = C_4$ acting on the disk D^2

$\tilde{H}_*(D^2) = 0$, so X is acyclic.

Fixed set: a single point, the center, so $\tilde{H}_*(X^G) = 0$.

SMITH THEORY

Smith's theorem for acyclic spaces

SMITH, 1938 TO 1941

A finite 2-group G acts on a sufficiently nice space X. Then

$$\tilde{H}_*(X) = 0 \quad \Longrightarrow \quad \tilde{H}_*(X^G) = 0.$$

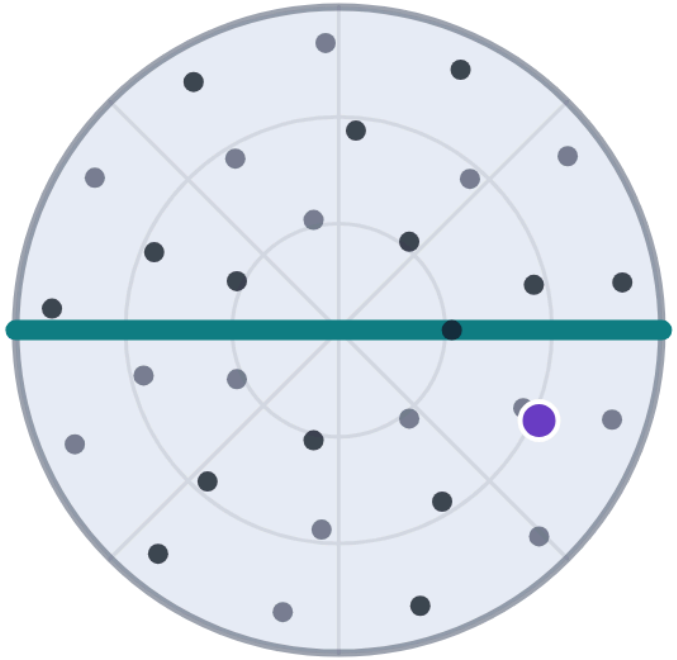
In particular $X^G \neq \emptyset$.

Example. $G = C_4$ acting on the disk D^2

$\tilde{H}_*(D^2) = 0$, so X is acyclic.

Fixed set: a segment, a diameter, so $\tilde{H}_*(X^G) = 0$.

Quarter turn | Reflection



MY RESEARCH, WITH NICHOLAS J. KUHN

Why a chromatic setting?

Algebraic topology always wants better, more refined invariants.

At the prime 2, the **Morava K-theories** $K(n)$ interpolate between rational homology and mod 2 homology.

$K(0)$	$K(1)$	$K(2), K(3), \ldots$	$K(\infty)$
rational homology	related to K-theory of vector bundles	everything in between	ordinary mod 2 homology

Each height sees at least as much as the one before: $k_0(X) \leq k_1(X) \leq \cdots \leq k_\infty(X)$.

Think of a prism splitting light into colours: each level shows a different layer of the same space. Hence **chromatic**.

MY RESEARCH, WITH NICHOLAS J. KUHN

The chromatic setting

Smith theory lives at the top height, $n = \infty$. We can ask the same question at every height $K(n)$. Write $k_n(X) = \dim K(n)_*(X)$.

Symmetry moves us between heights: the fixed points are seen at a lower height than the space. How far down?

CHROMATIC SMITH (G, n, m)

$$\tilde{K}(m)_*(X) = 0 \implies \tilde{K}(n)_*(X^G) = 0$$

CHROMATIC FLOYD (G, n, m)

$$k_m(X) \geq k_n(X^G)$$

For every sufficiently nice G-space X, with G a finite 2-group. The height can only drop going to the fixed points: from m down to n.

MY RESEARCH, WITH NICHOLAS J. KUHN

Chromatic Smith implies chromatic Floyd

	Classical (mod 2 homology)	Chromatic (Morava K-theory)
Floyd $\Rightarrow$ Smith	**Known.** A short argument.	**Immediate** from the definitions.
Smith $\Rightarrow$ Floyd	**Now a consequence** of the chromatic theorem.	`Theorem, Kuhn–Lloyd` Chromatic Smith implies chromatic Floyd. The two are equivalent.

Idea. If Floyd fails, take a space X that violates it. Cut a summand out of $X^{\wedge k}$ using an idempotent in $\mathbb{Z}_{(2)}[\Sigma_k]$. That summand violates Smith. (An idea of J. Smith, used by Hopkins and Smith.)

N. J. Kuhn and C. J. R. Lloyd, Chromatic fixed point theory and the Balmer spectrum for extraspecial 2-groups, Amer. J. Math. **146** (2024), 769–812.

MY RESEARCH, WITH NICHOLAS J. KUHN

The blue shift number

BLUE SHIFT NUMBER

$r_n(G)$ is the smallest r such that for every sufficiently nice G-space X,

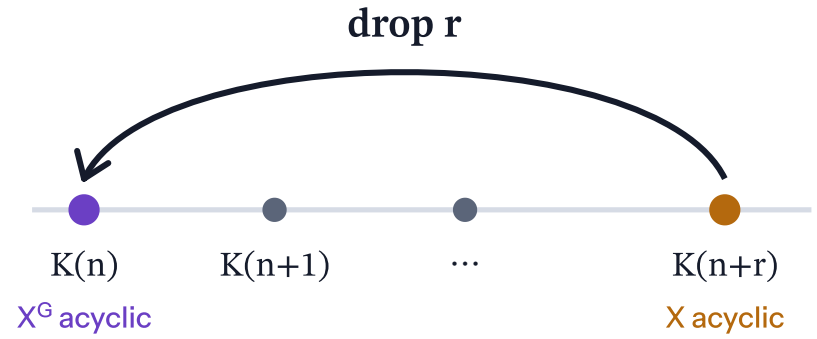
$$\tilde{K}(n+r)_*(X) = 0 \implies \tilde{K}(n)_*(X^G) = 0.$$

It measures the drop in height from X to its fixed points X^G.

Part of the game: determine the minimum drop in height.

CONJECTURE (KUHN–LLOYD)

$r_n(G) = r_+(G)$ for every finite 2-group G and height n. Here $r_+(G)$ is the length of the shortest chain $1 = K_0 \trianglelefteq K_1 \trianglelefteq \cdots \trianglelefteq K_r = G$ with every quotient K_i/K_{i-1} cyclic: the cyclic length of G. In particular, the drop does not depend on n.



Always $0 \le r_n(G) \le \log_2 |G|$.

C_2 **acting:** $r = 1$ (Strickland; Balmer–Sanders).

Abelian G**:** $r = \mathrm{rank}(G)$ (Barthel et al.).

Extraspecial 2-groups: determined by Kuhn–Lloyd.

Resolved

OpenAI has posted a preprint proving it: $r_n(G) = r_+(G)$ for every finite 2-group and every height.

Cyclic length and chromatic fixed-point loss, OpenAI Preprint #314, September 2026.

MY RESEARCH

Using the general theory

Now that we have this general theory, we can use it to calculate the Morava K-theory of spaces.

1. Blue shift lower bound

If a cyclic 2-group C acts on X, the blue shift bound gives $k_n(X) \geq k_{n-1}(X^C)$.

2. Spectral sequence upper bound

The Atiyah–Hirzebruch spectral sequence gives $k_n(X) \leq \dim H(X; Q_n)$, the Margolis homology of the nth Milnor operation Q_n.

3. They meet

Then $k_n(X)$ is known exactly, and the spectral sequence collapses.

I searched for group actions on real Grassmannians whose fixed points make the bounds meet.

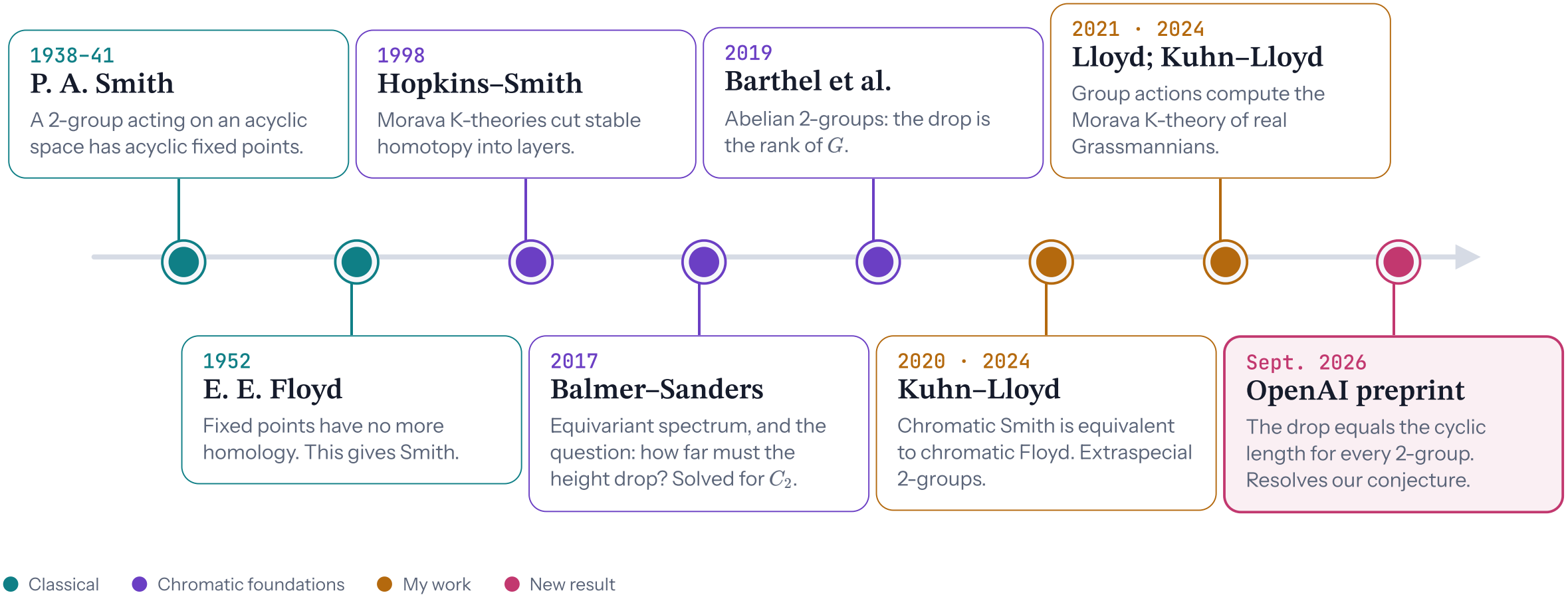
This was my dissertation. It computes $k_n(\mathrm{Gr}_d(\mathbb{R}^m)) = \binom{m}{d}$ for $m \leq 2^{n+1}$, and $k_n(\mathrm{Gr}_2(\mathbb{R}^m))$ for all m.

This involved writing **Python** and **GAP** code to search for these actions using representation theory.

C. J. R. Lloyd, Applications of chromatic fixed point theory, PhD thesis, University of Virginia, 2021.
N. J. Kuhn and C. J. R. Lloyd, Computing the Morava K-theory of real Grassmannians using chromatic fixed point theory, Algebr. Geom. Topol. **24** (2024), 919–950.

SUMMARY

The story so far



NEW SECTION

Topology in Aviation

I bring my topological taste to work.

TOPOLOGY IN AVIATION

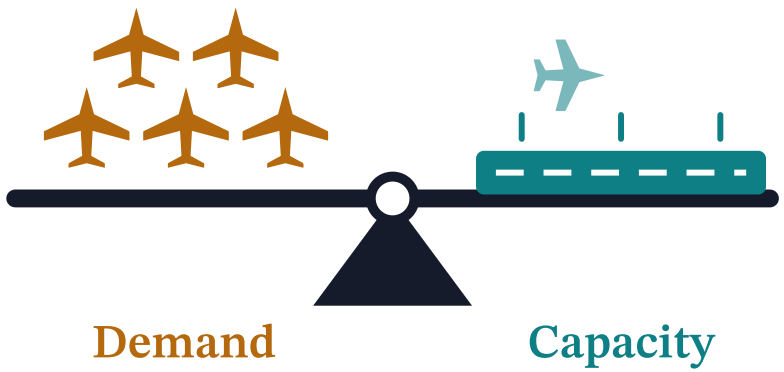
The setting

The command center **balances demand and capacity** in the National Airspace System.

The **COO** wanted the command center to be able to detect **airborne holding** in near real time.

My group, the **product development group**, was tasked with building the tool.

I built the entire tool from the ground up.



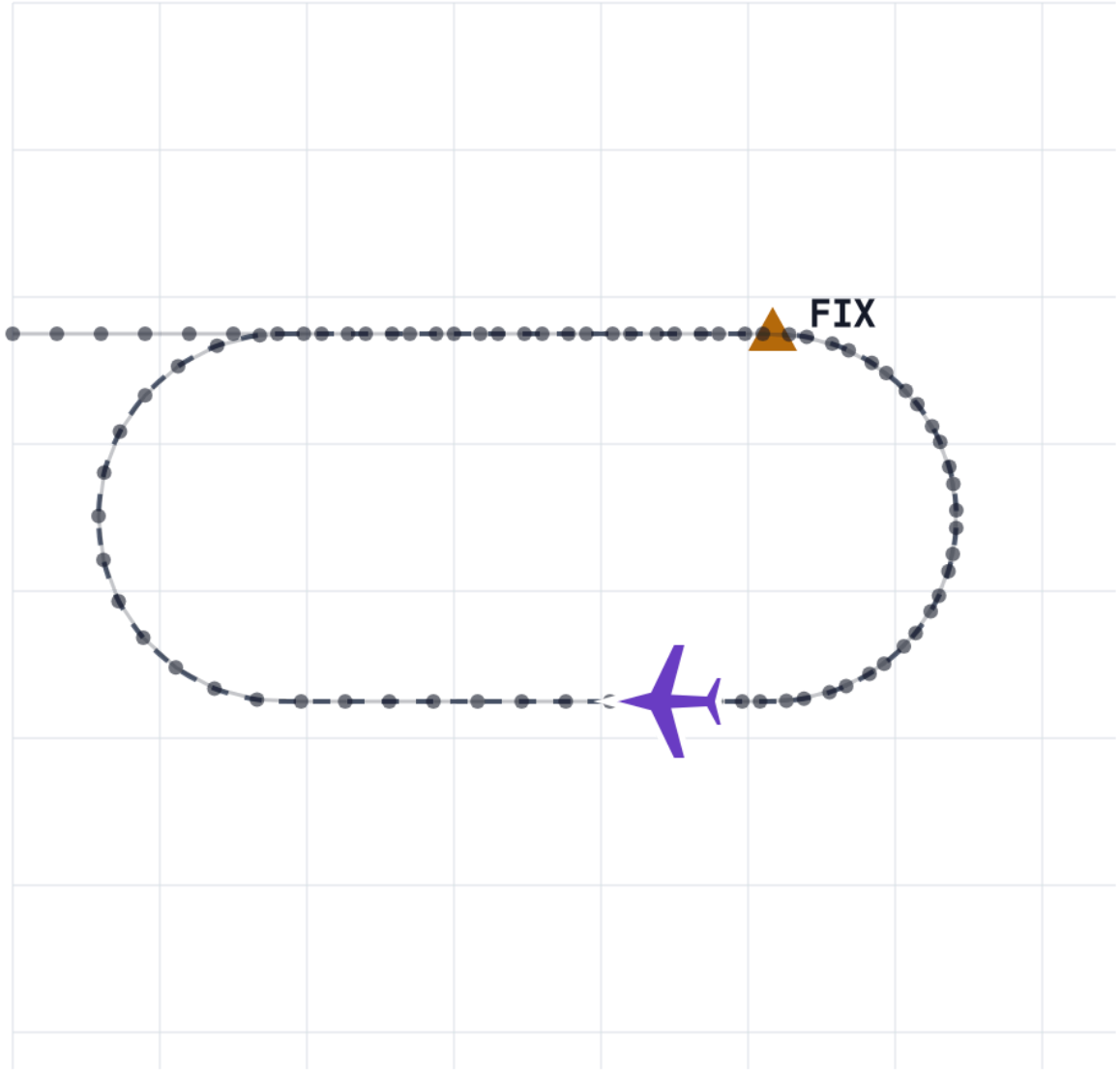
TOPOLOGY IN AVIATION

The problem: airborne holding

Air traffic control may **hold** an aircraft in a racetrack pattern over a fix while it waits for clearance.

The pattern: a straight leg to the fix, a turn, a leg back, a turn, and again.

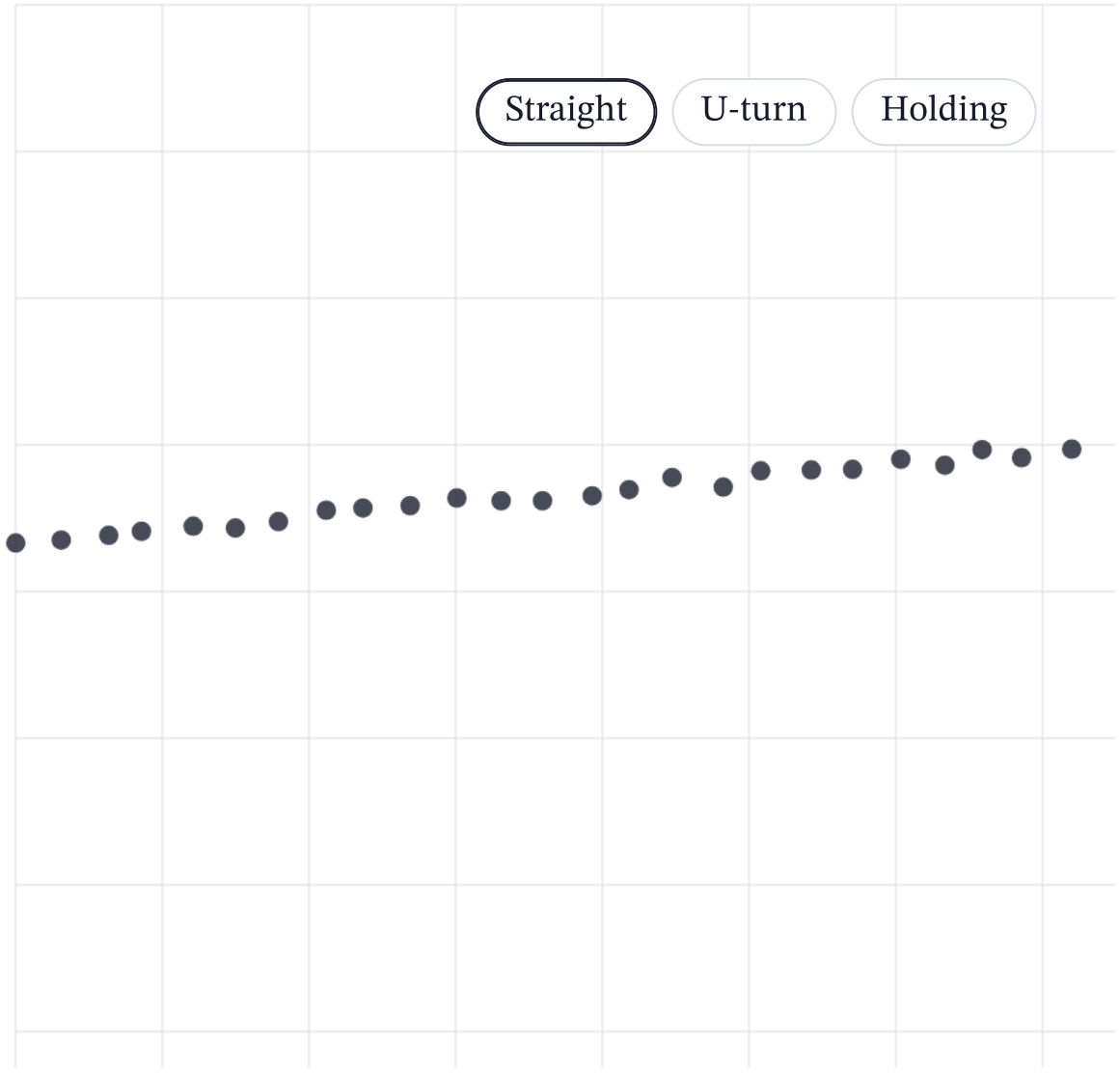
Problem: from the radar track alone, decide when an aircraft was holding.



TOPOLOGY IN AVIATION

How would we detect it?

A radar track is a list of timestamped positions.



TOPOLOGY IN AVIATION

How would we detect it?

A radar track is a list of timestamped positions.

We want a test that is quick and survives noise.



radar returns: 39 timestamped positions

TOPOLOGY IN AVIATION

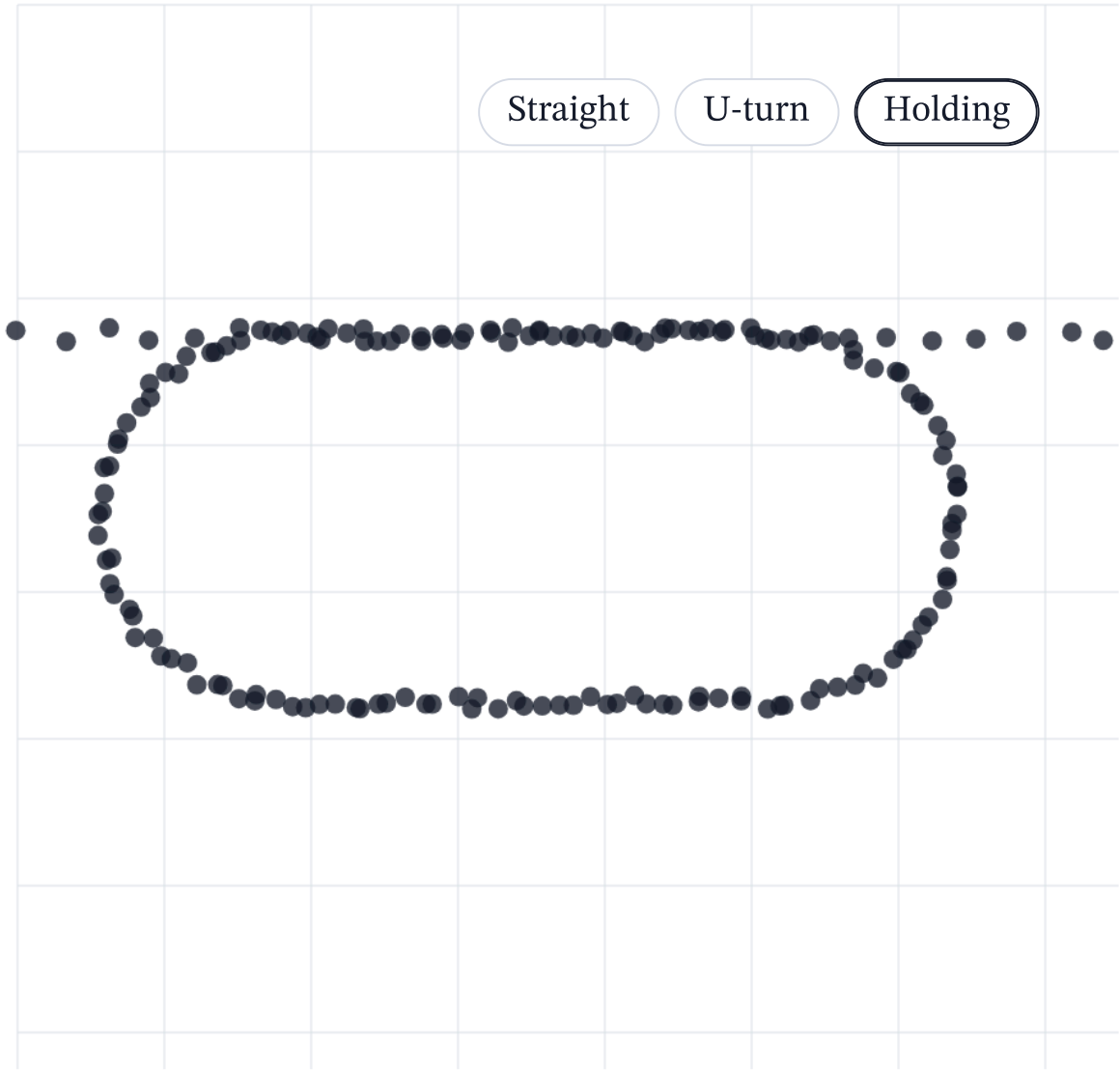
How would we detect it?

A radar track is a list of timestamped positions.

We want a test that is quick and survives noise.

Let's model it topologically.

Treat the track as a shape, and ask what shape it is.



radar returns: 171 timestamped positions

TOPOLOGY IN AVIATION

Modeling the track topologically

MOTIVATION: TRANSVERSALITY (THOM, 1954)

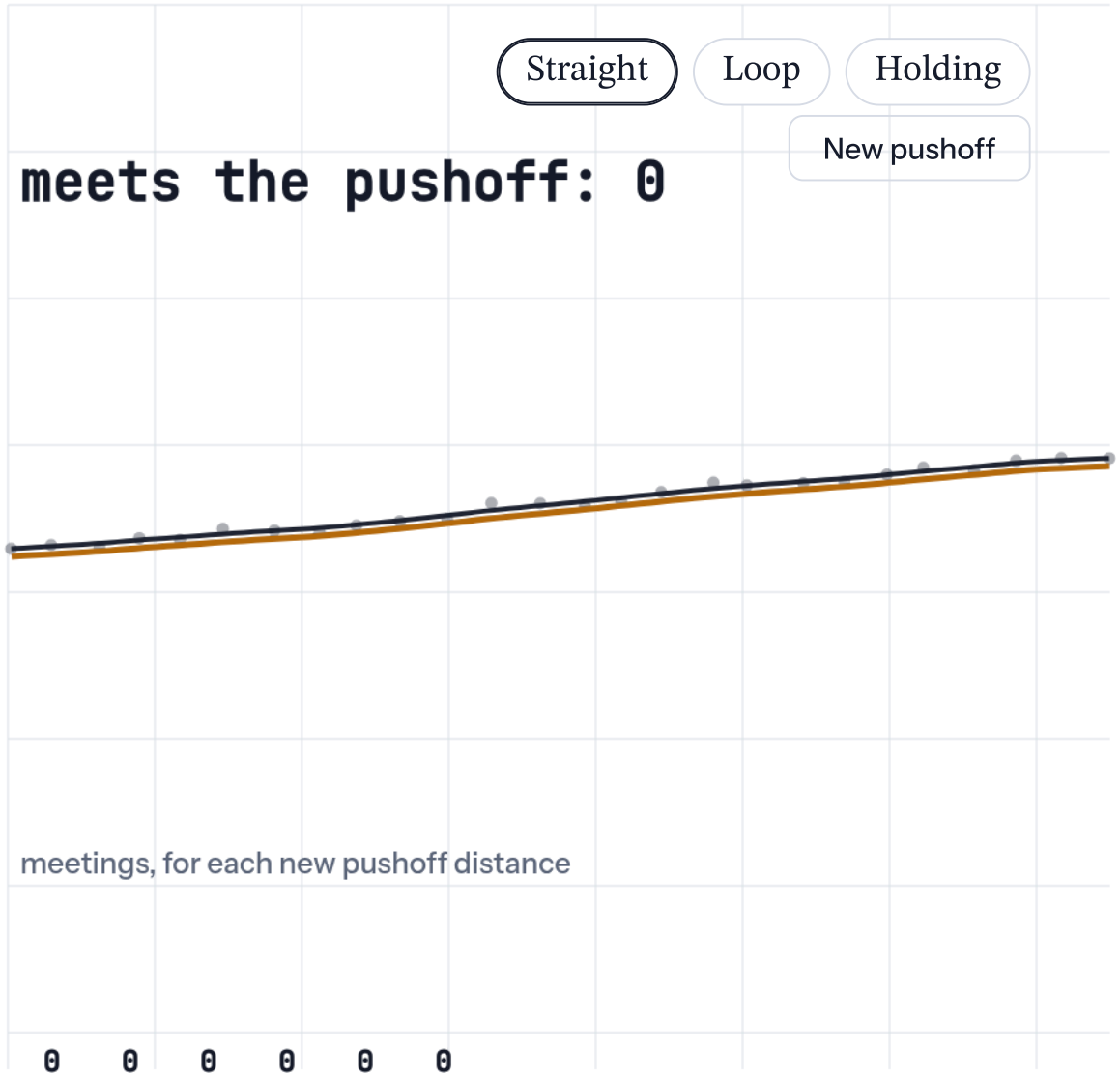
A generic curve crosses itself only **transversely** (at a genuine angle, not touching along a tangent), and transverse crossings survive small pushoffs. Holding lies on top of itself, so it is not generic: its pushoff meetings come and go.

1. Smooth the track into a linestring.

2. Treat it as a curve in the plane. Is it an embedding? Airborne holding crosses itself.

3. Copy the curve and push it off by a constant distance. Count where the copy meets the original. Is the count stable as the distance changes?

Stable: always 0.



R. Thom, Quelques propriétés globales des variétés différentiables, Comment. Math. Helv. **28** (1954), 17–86.
See also Guillemin–Pollack, Differential Topology, Ch. 2.

TOPOLOGY IN AVIATION

Modeling the track topologically

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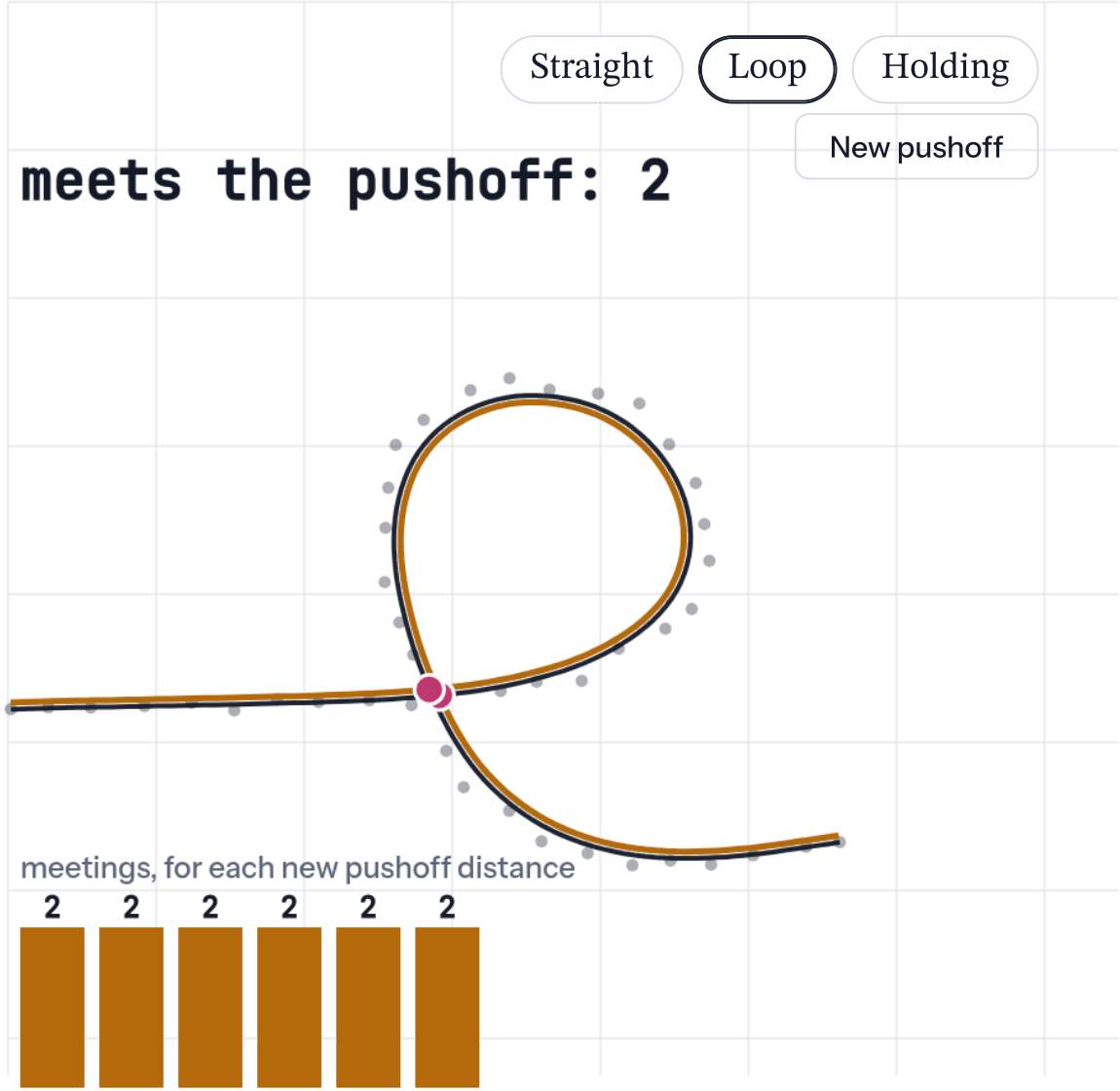
A generic curve crosses itself only **transversely** (at a genuine angle, not touching along a tangent), and transverse crossings survive small pushoffs. Holding lies on top of itself, so it is not generic: its pushoff meetings come and go.

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Stable: always 2.



R. Thom, Quelques propriétés globales des variétés différentiables, Comment. Math. Helv. **28** (1954), 17–86.
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TOPOLOGY IN AVIATION

Modeling the track topologically

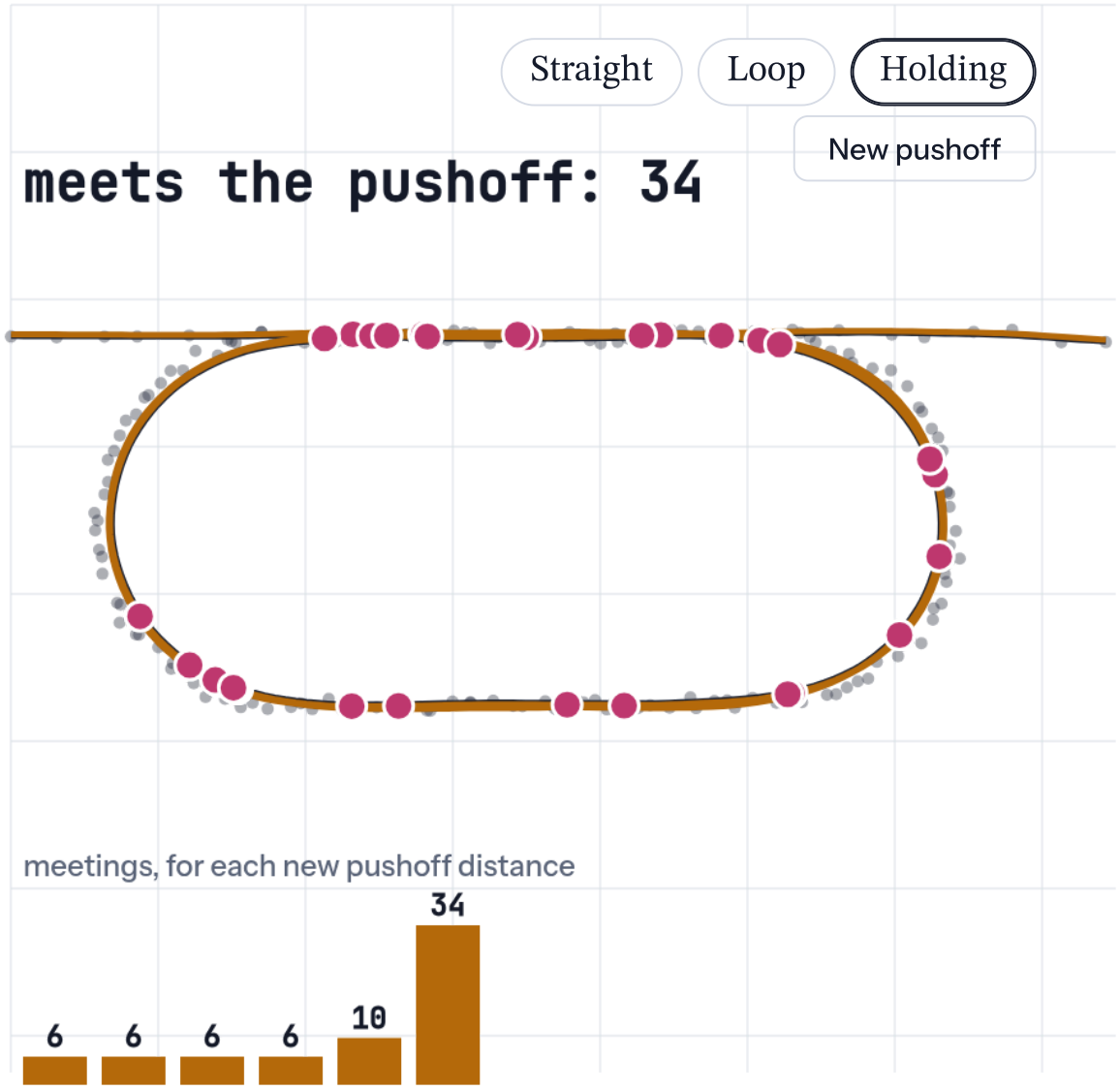
> MOTIVATION: TRANSVERSALITY (THOM, 1954)
> A generic curve crosses itself only **transversely** (at a genuine angle, not touching along a tangent), and transverse crossings survive small pushoffs. Holding lies on top of itself, so it is not generic: its pushoff meetings come and go.

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3. Copy the curve and push it off by a constant distance. Count where the copy meets the original. Is the count stable as the distance changes?

Unstable: between 6 and 34.



R. Thom, Quelques propriétés globales des variétés différentiables, Comment. Math. Helv. **28** (1954), 17–86.
See also Guillemin–Pollack, Differential Topology, Ch. 2.

Implementation and validation

Implementation. The algorithm is written entirely in PostGIS, as spatial SQL.

Validation. I validated the output against next day airborne holding reporting.

Each flight lands in one of three buckets: flagged by both, flagged only by the algorithm, or reported only.

In production: the Near Real Time tool

I used this algorithm to detect airborne holding in near real time.

Deployed into a production tool, the **Near Real Time tool**, used down at the command center.

The tool also shows other delay metrics that we calculate, such as **surface delay** and **program delay**, across the entire **National Airspace System**.

Thank you!

Questions?

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Kuhn–Lloyd and my work

N. J. Kuhn and C. J. R. Lloyd, *Chromatic fixed point theory and the Balmer spectrum for extraspecial 2-groups*, Amer. J. Math. **146** (2024), 769–812. arXiv:2008.00330.

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Recent

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